CAPSTONE PROJECT 3: SIEM DEPLOYMENT

1. Executive Summary

This project involved the deployment of a Security Information and Event Management (SIEM) solution using Splunk to provide centralized monitoring and security visibility for an organization's IT infrastructure. An Ubuntu Server was configured as the central Splunk server, while Windows and Linux systems were connected as monitored endpoints using Splunk Universal Forwarders.

The project focused on collecting and analysing system and security logs from the endpoints. Four controlled attack scenarios were simulated: brute-force login attempts, unauthorized account creation, suspicious command execution, and clearing Windows Security logs. The resulting events were collected by Splunk and investigated to determine how the attacks could be detected.

The project demonstrated the importance of centralized logging in a **Security Operations Centre (SOC)**. Splunk provided a central location for monitoring events, investigating suspicious activity, identifying attack timelines, and supporting incident response.

2. Project Objectives

The main objectives of the project were to:

- Deploy a central Splunk SIEM server.
- Connect Windows and Linux endpoints to Splunk.
- Collect Windows System, Application, and Security logs.
- Collect Linux authentication and system logs.
- Simulate different cybersecurity attacks in a controlled lab setup.
- Detect and investigate the simulated attacks using Splunk.
- Map identified attacks to MITRE ATT&CK techniques.
- Provide recommendations for preventing similar incidents.

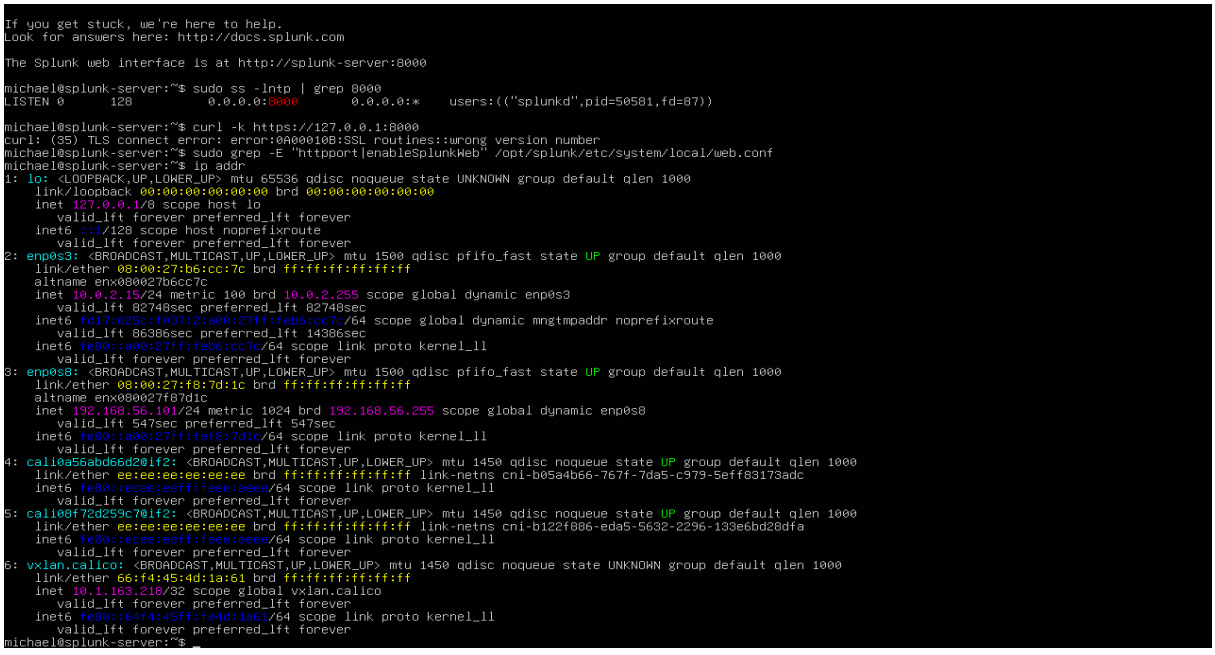
3. Lab Setup / Infrastructure

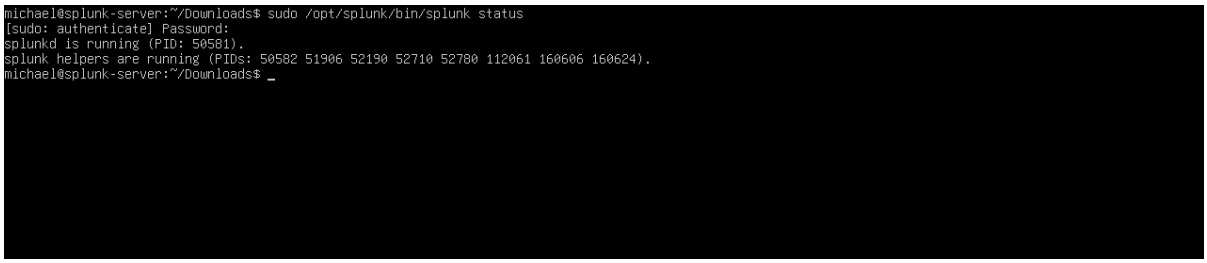
The project used a virtualized cybersecurity lab consisting of a central Ubuntu Server, a Windows endpoint, and Linux endpoint which are used for controlled security testing.

SYSTEM	ROLE	IP ADDRESS
Ubuntu Server	Splunk Siem Server	198.168.56.101
Windows	Monitored Endpoints	10.95.216.214
Linux	Monitored Endpoints	10.0.2.15

All systems were connected to the same virtual network to allow communication and logs forwarding.

Splunk Enterprise was installed on the Ubuntu Server. Splunk Universal Forwarder was installed on the Windows and Linux endpoints to forward relevant logs to the central Splunk server.





```
michael@splunk-server:~$ sudo /opt/splunk/bin/splunk restart
Running Splunk Enterprise as root is deprecated and will be removed in a future release. For details, see the Release Notes. To run as root, use the --run-as-ro
ot option.
michael@splunk-server:~$ sudo /opt/splunk/bin/splunk restart --run-as-root
Stopping splunkd...
Shutting down.  Please wait, as this may take a few minutes.
....
Stopping splunk helpers...

Done.

Splunk> All batbelt. No tights.

Checking prerequisites...
        Checking http port [8000]: open
        Checking mgmt port [8089]: open
        Checking appserver port [127.0.0.1:8065]: open
        Checking kvstore port [8191]: open
        Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
        Checking critical directories...        Done
        Checking indexes...
                Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_rollup _telemetry _th
efishbucket history main summary
        Done
        Checking filesystem compatibility...  Done
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from '/opt/splunk/splunk-10.6.0.5-86587d4e3b27-linux-amd64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done


Waiting for web server at http://127.0.0.1:8000 to be available..................................... Done


If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://splunk-server:8000

michael@splunk-server:~$ sudo ss -lntp |grep 9997
LISTEN 0      128          0.0.0.0:9997       0.0.0.0:*    users:(("splunkd",pid=197110,fd=70))

michael@splunk-server:~$
```

4. Splunk Server Deployment

The central Splunk server was deployed on Ubuntu Server. Splunk Enterprise was installed and configured to receive data from the connected endpoints.

The Splunk receiving port was configured as: **TCP 9997**

Splunk Web was accessed through: **Port 8000**

```
Stopping splunk helpers...

Done.
Stopped helpers.
Removing stale pid file... done.
splunkd is not running.

Splunk> 4TW

Checking prerequisites...
        Checking http port [8000]: open
        Checking mgmt port [8089]: open
        Checking appserver port [127.0.0.1:8065]: open
        Checking kvstore port [8191]: open
        Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
        Checking critical directories...        Done
        Checking indexes...
                Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_rollup _telemetry _th
efishbucket history main summary windows
        Done
        Checking filesystem compatibility...  Done
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from '/opt/splunk/splunk-10.6.0.5-86587d4e3b27-linux-amd64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done


Waiting for web server at http://127.0.0.1:8000 to be available..................................................... Done


If you get stuck, we're here to help.
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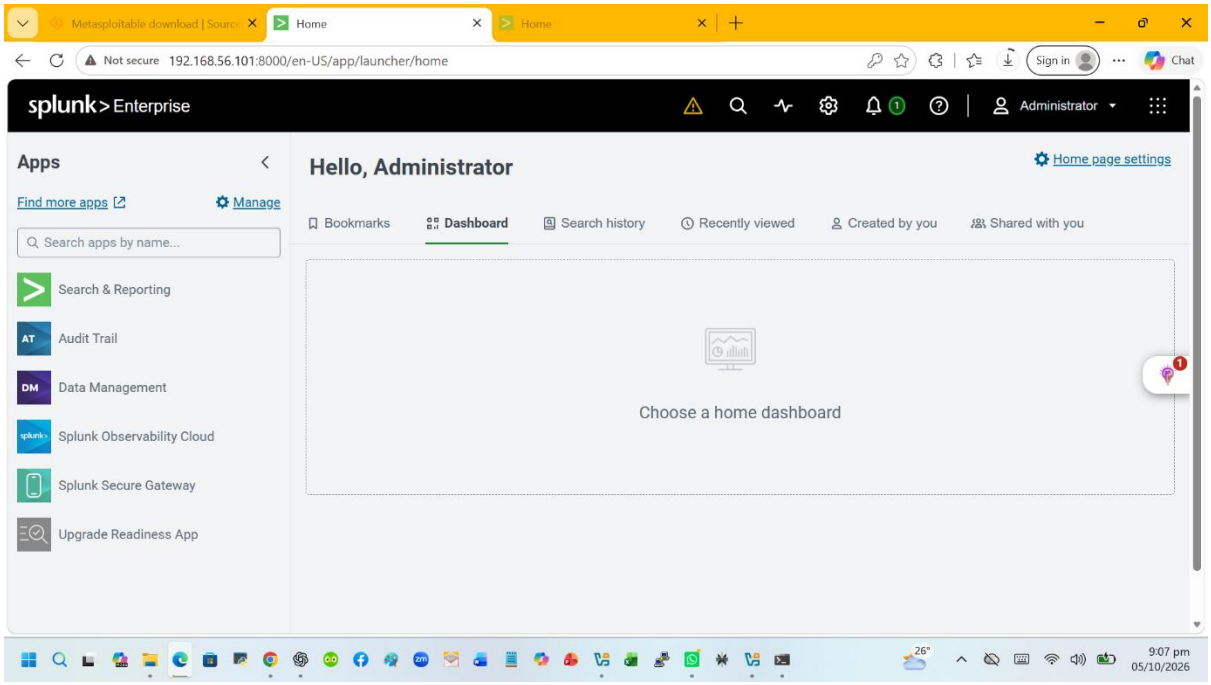
The Splunk web interface is at http://splunk-server:8000

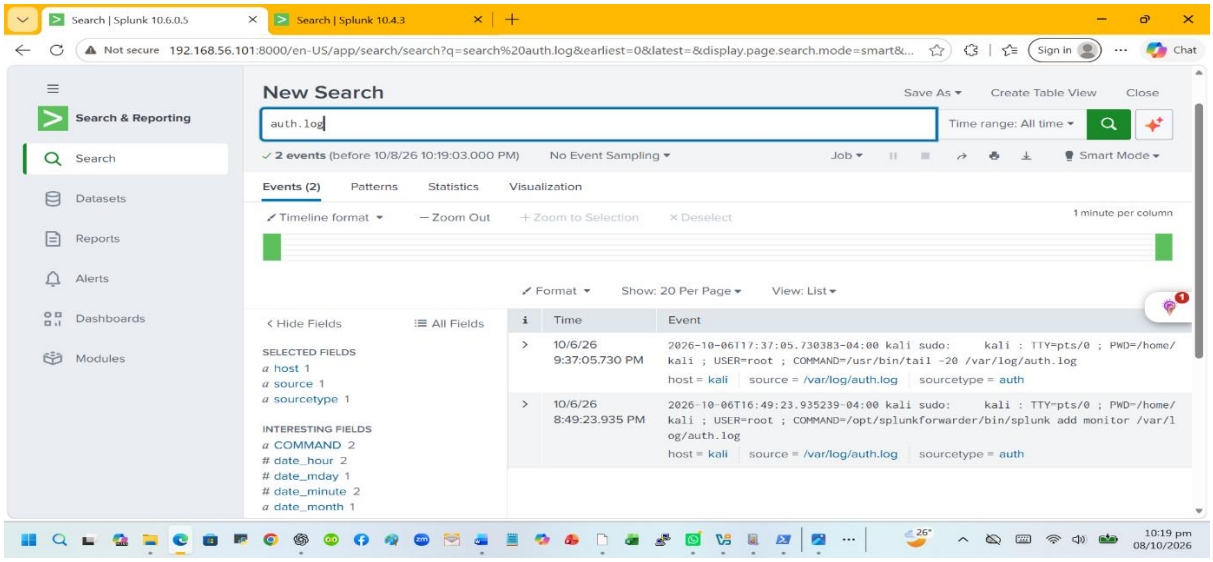
michael@splunk-server:~$ sudo ss -lntp | grep 8000
[sudo: authenticate] Password:
LISTEN 0      128          0.0.0.0:8000       0.0.0.0:*    users:(("splunkd",pid=36071,fd=192))

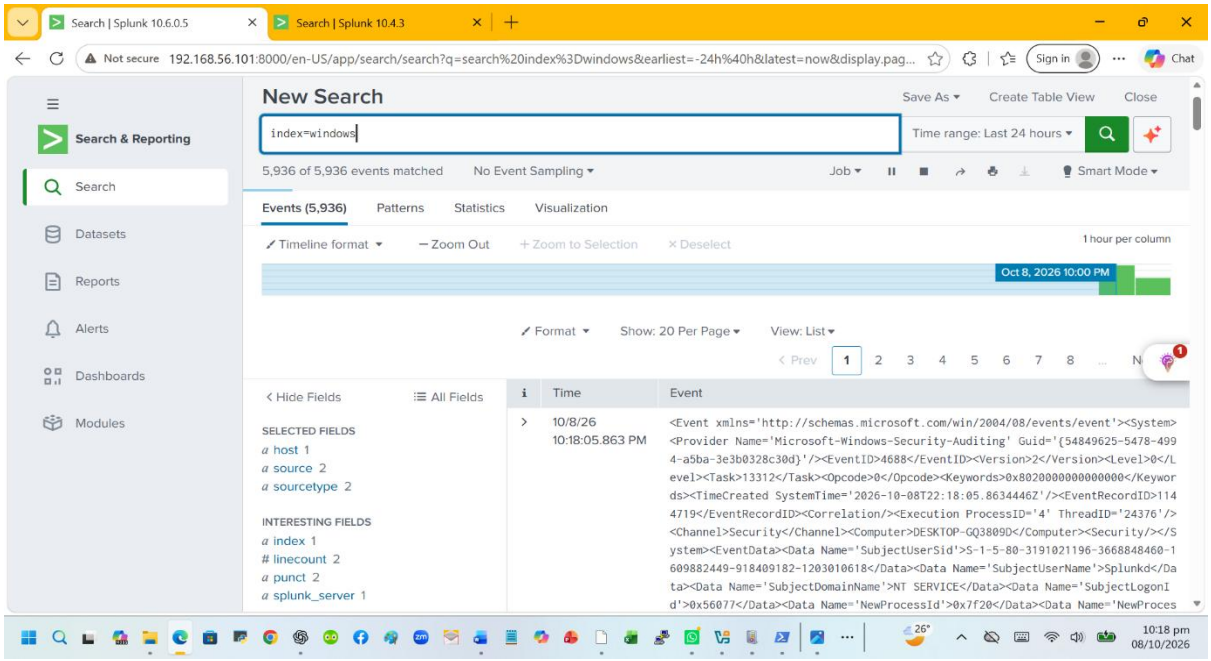
michael@splunk-server:~$ sudo ss -lntp | grep 9997
LISTEN 0      128          0.0.0.0:9997       0.0.0.0:*    users:(("splunkd",pid=36071,fd=74))

michael@splunk-server:~$ _
```

The server was tested to ensure that it could communicate with the Windows and Linux endpoints and receive forwarded security events.







5. Windows Logs Collection

The Windows endpoint was configured with Splunk Universal Forwarder. The following Windows logs were monitored:

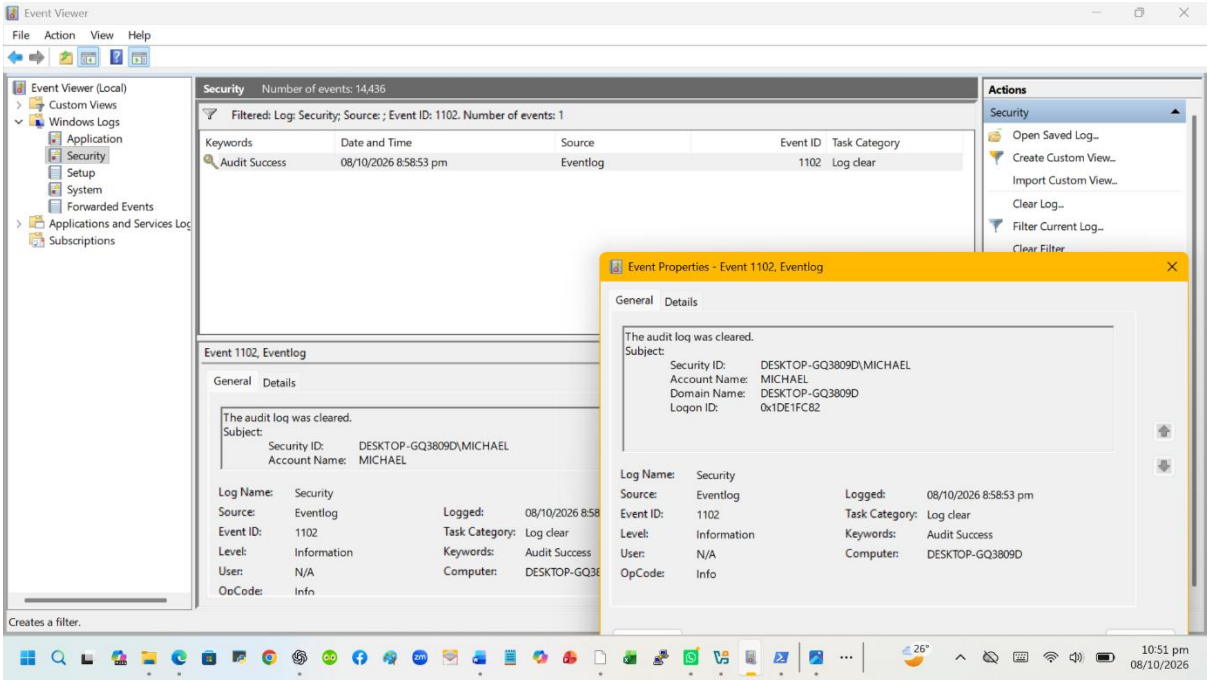
- **Application**
- **Security**
- **System**

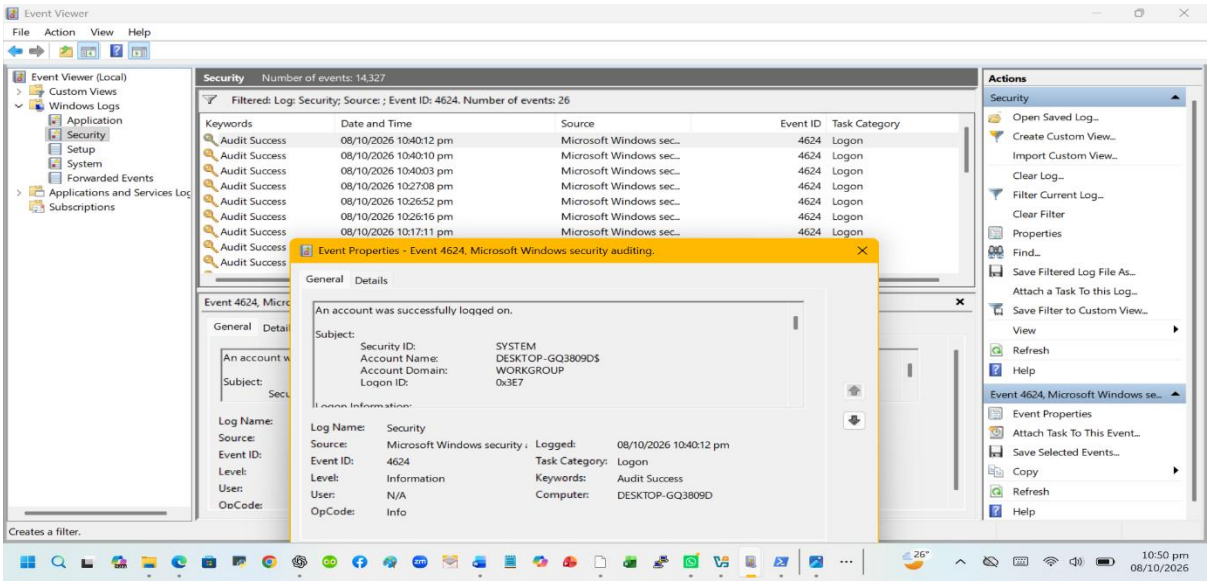
The logs were forwarded to the central Splunk server for analysis.

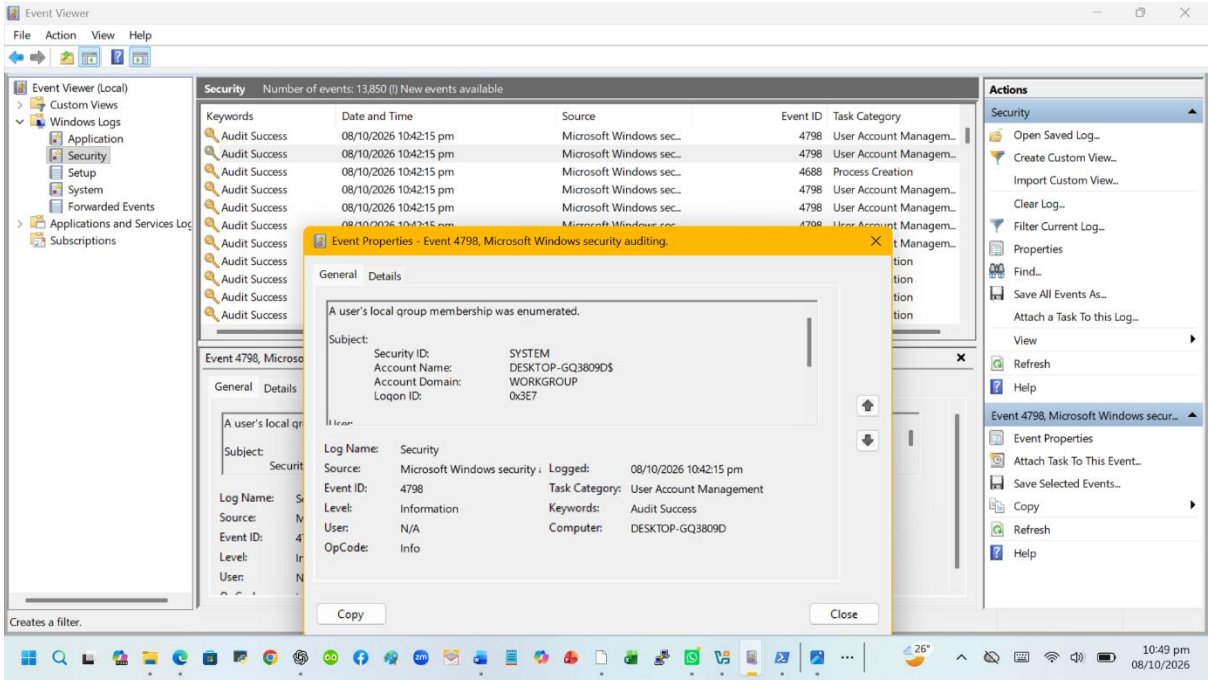
The Windows Security log was particularly important because it contains events relating to authentication, account creation, security changes, and other security-related activities.

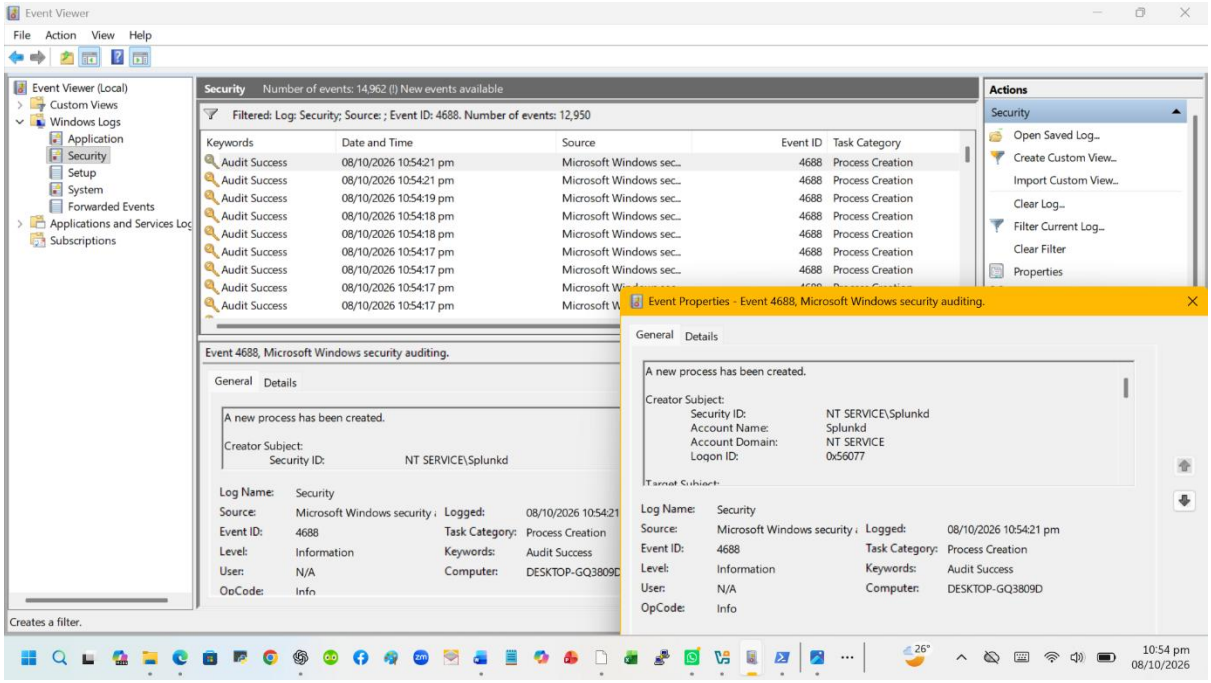
Examples of important Windows events included:

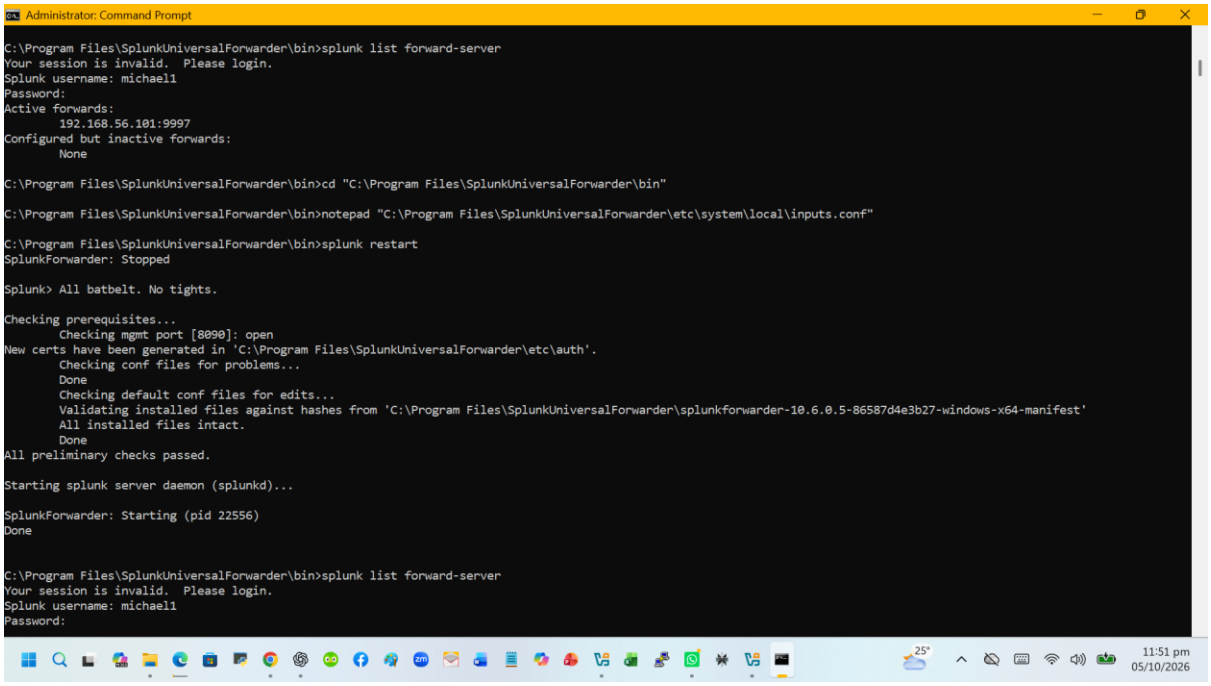
- **4624** – Successful logon
- **4798** – User account created
- **1102** – Security audit log cleared
- **4688** – Process creation, when appropriate auditing is enabled

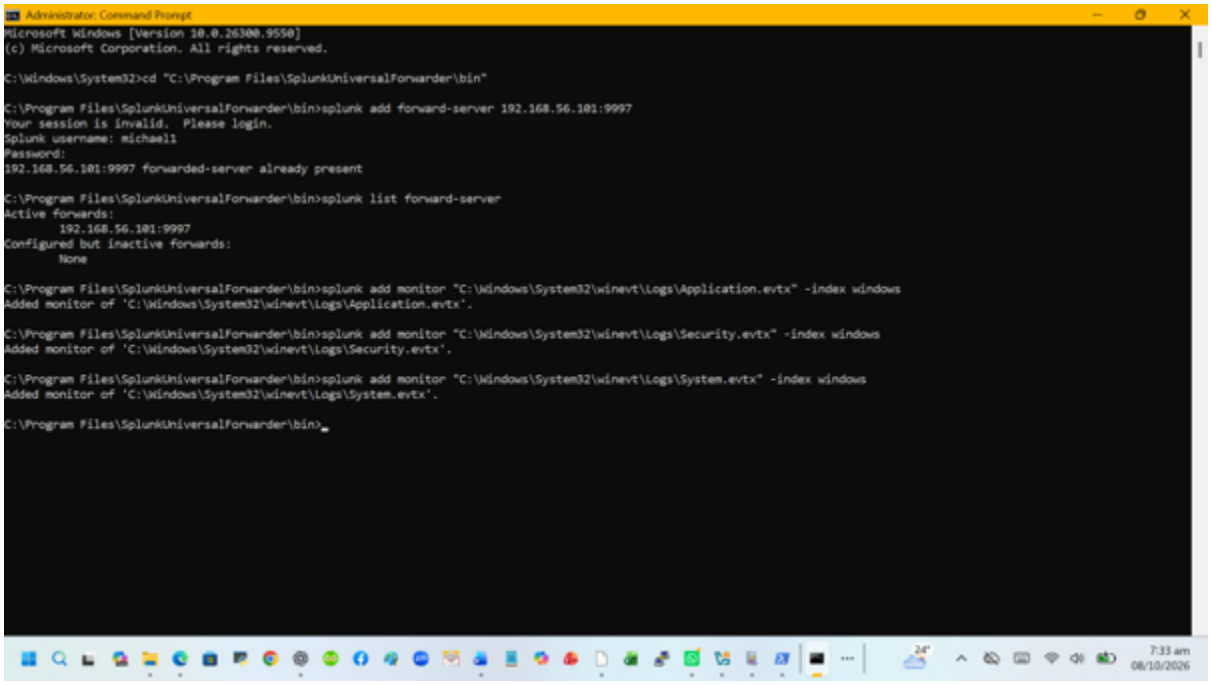


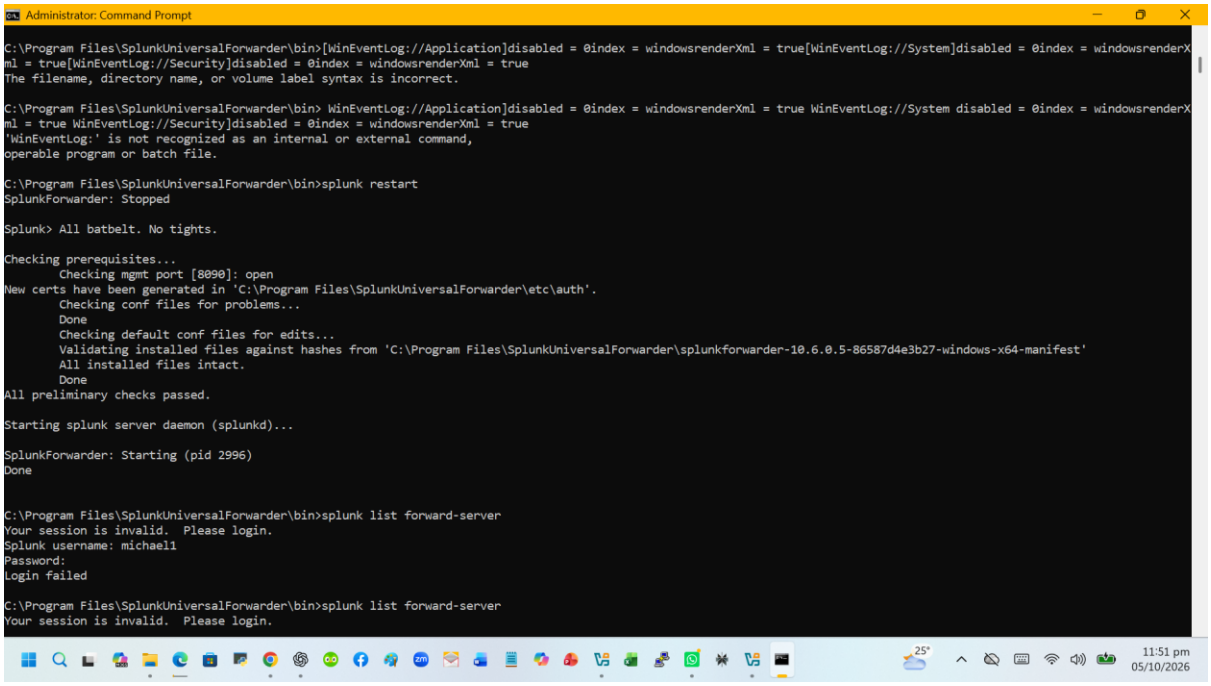


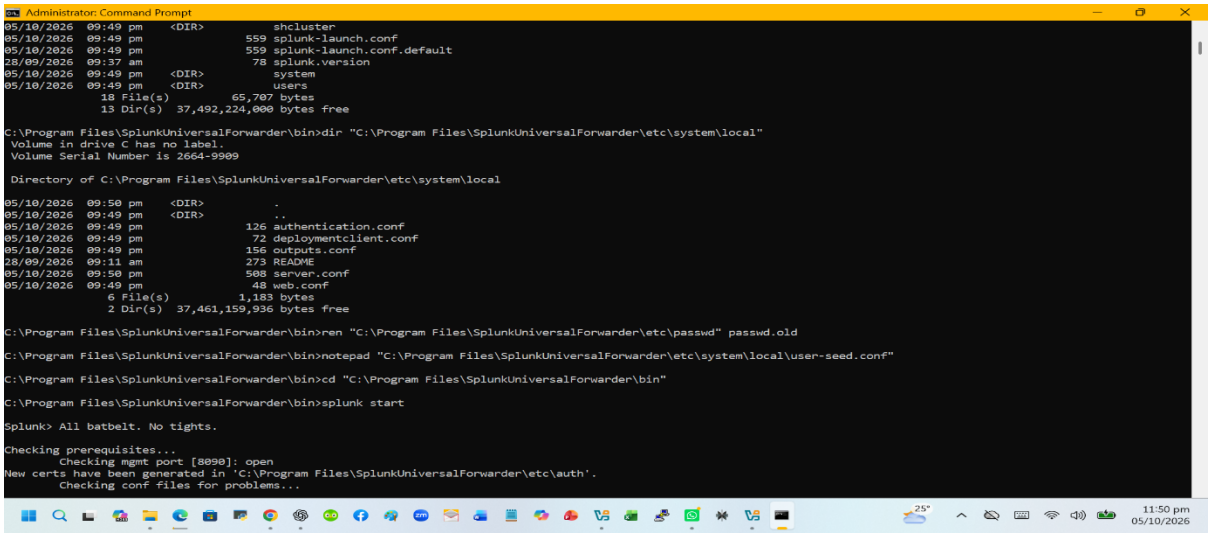

Event Viewer
File Action View Help
Event Viewer (Local)
Custom Views
Windows Logs
Application
Security
Setup
System
Forwarded Events
Applications and Services Log
Subscriptions
Security Number of events: 13,850 (!) New events available
Keywords Date and Time Source Event ID Task Category
Audit Success 08/10/2026 10:42:15 pm Microsoft Windows sec... 4798 User Account Managem...
Audit Success 08/10/2026 10:42:15 pm Microsoft Windows sec... 4798 User Account Managem...
Audit Success 08/10/2026 10:42:15 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:42:15 pm Microsoft Windows sec... 4798 User Account Managem...
Audit Success 08/10/2026 10:42:15 pm Microsoft Windows sec... 4798 User Account Managem...
Event Properties - Event 4798, Microsoft Windows security auditing.
General Details
A user's local group membership was enumerated.
Subject:
Security ID: SYSTEM
Account Name: DESKTOP-GQ3809D$
Account Domain: WORKGROUP
Logon ID: 0x3E7
Log Name: Security
Source: Microsoft Windows security Logged: 08/10/2026 10:42:15 pm
Event ID: 4798 Task Category: User Account Management
Level: Information Keywords: Audit Success
User: N/A Computer: DESKTOP-GQ3809D
OpCode: Info
Copy
Close
Actions
Security
Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Properties
Find...
Save All Events As...
Attach a Task To this Log...
View
Refresh
Help
Event 4798, Microsoft Windows secur...
Event Properties
Attach Task To This Event...
Save Selected Events...
Copy
Refresh
Help
Creates a filter.
10:49 pm
08/10/2026

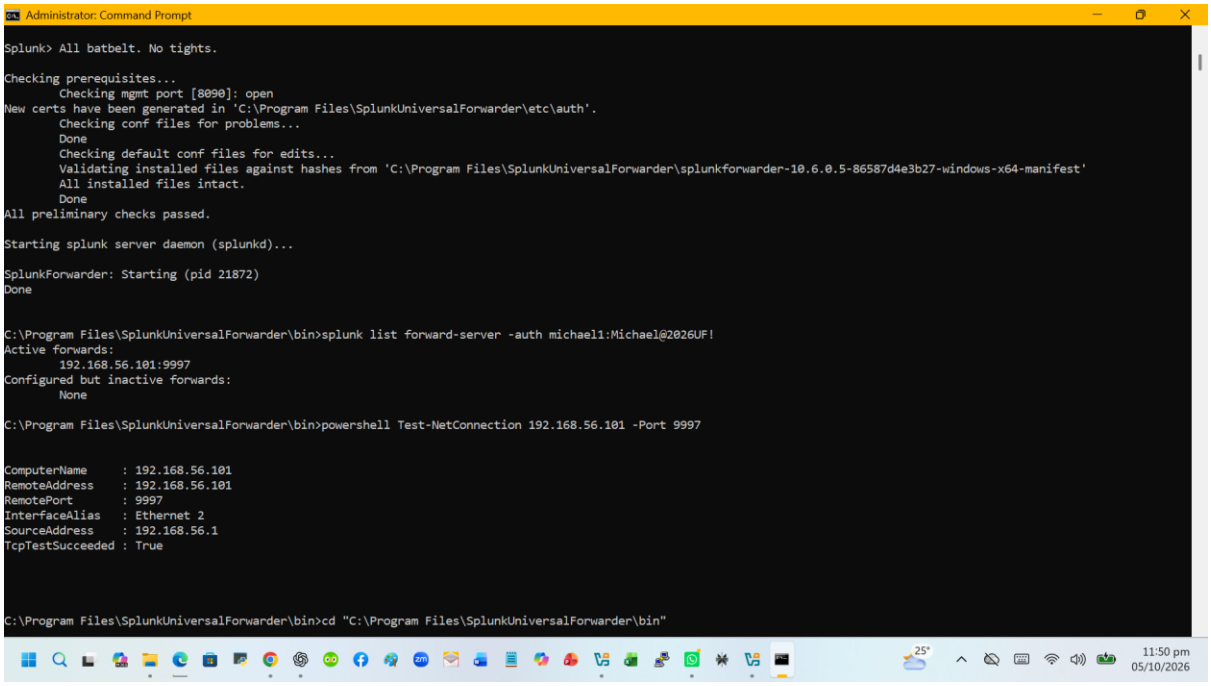

Event Viewer
File Action View Help
Event Viewer (Local)
Custom Views
Windows Logs
Application
Security
Setup
System
Forwarded Events
Applications and Services Log
Subscriptions
Security Number of events: 14,962 (!) New events available
Filtered: Log: Security; Source: ; Event ID: 4688. Number of events: 12,950
Keywords Date and Time Source Event ID Task Category
Audit Success 08/10/2026 10:54:21 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:21 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:19 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:18 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:18 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:17 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:17 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:17 pm Microsoft Windows sec... 4688 Process Creation
Audit Success 08/10/2026 10:54:17 pm Microsoft Windows sec...
Event 4688, Microsoft Windows security auditing.
General Details
A new process has been created.
Creator Subject:
Security ID: NT SERVICE\Splunkd
Log Name: Security
Source: Microsoft Windows security Logged: 08/10/2026 10:54:21
Event ID: 4688 Task Category: Process Creation
Level: Information Keywords: Audit Success
User: N/A Computer: DESKTOP-GQ3809D
OpCode: Info
Event Properties - Event 4688, Microsoft Windows security auditing.
General Details
A new process has been created.
Creator Subject:
Security ID: NT SERVICE\Splunkd
Account Name: Splunkd
Account Domain: NT SERVICE
Logon ID: 0x56077
Log Name: Security
Source: Microsoft Windows security Logged: 08/10/2026 10:54:21 pm
Event ID: 4688 Task Category: Process Creation
Level: Information Keywords: Audit Success
User: N/A Computer: DESKTOP-GQ3809D
OpCode: Info
Actions
Security
Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Clear Filter
Properties
Creates a filter.
10:54 pm
08/10/2026

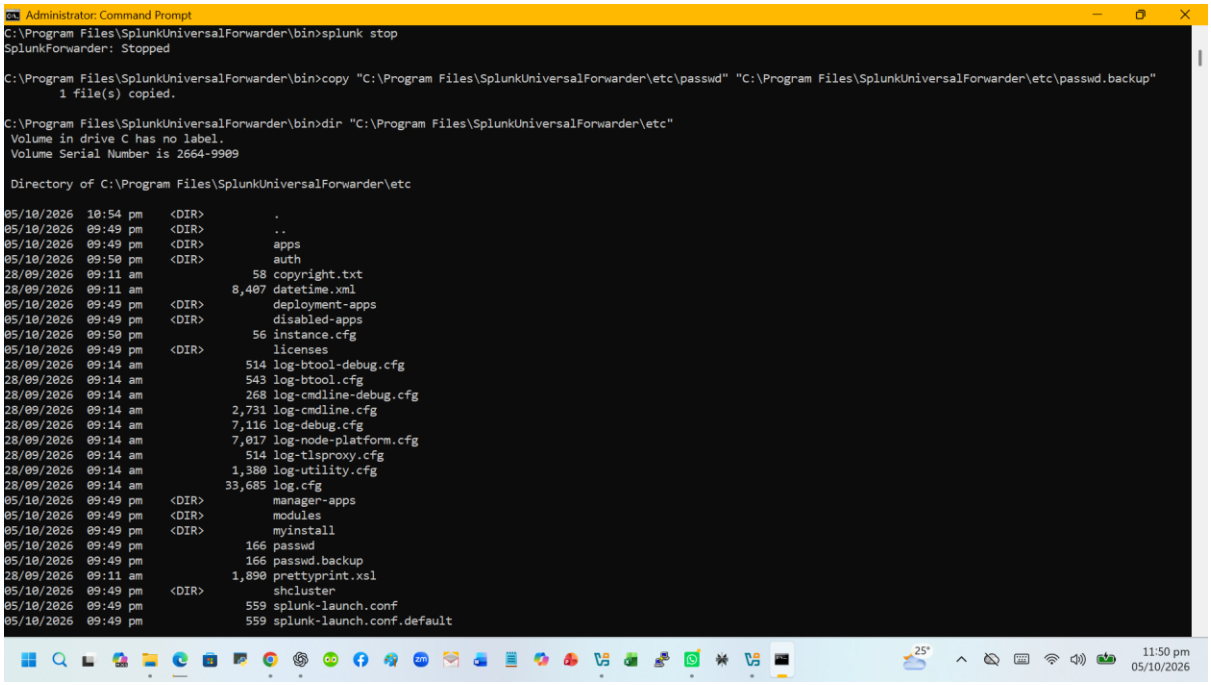

Administrator: Command Prompt
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
Splunk username: michael1
Password:
Active forwards:
192.168.56.101:9997
Configured but inactive forwards:
None
C:\Program Files\SplunkUniversalForwarder\bin>cd "C:\Program Files\SplunkUniversalForwarder\bin"
C:\Program Files\SplunkUniversalForwarder\bin>notepad "C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf"
C:\Program Files\SplunkUniversalForwarder\bin>splunk restart
SplunkForwarder: Stopped
Splunk> All batbelt. No tights.
Checking prerequisites...
Checking mgmt port [8090]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from 'C:\Program Files\SplunkUniversalForwarder\splunkforwarder-10.6.0.5-86587d4e3b27-windows-x64-manifest'
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
SplunkForwarder: Starting (pid 22556)
Done
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
Splunk username: michael1
Password:
25°
11:51 pm
05/10/2026

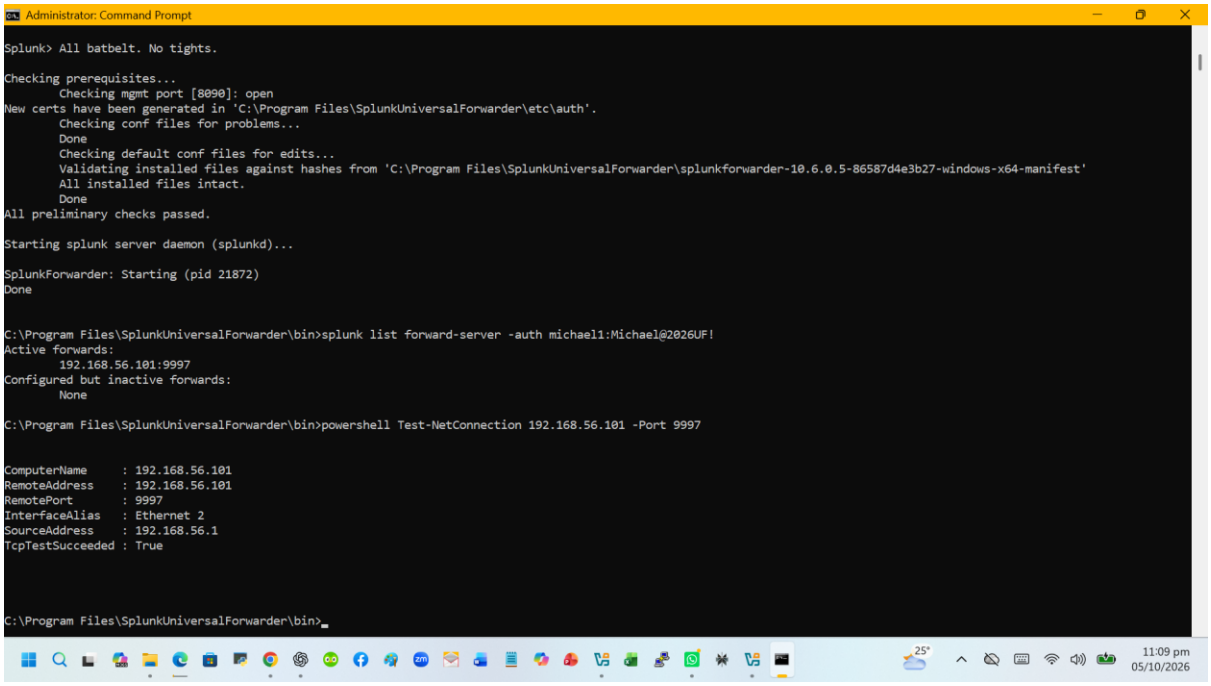

Administrator: Command Prompt
Microsoft Windows [Version 10.0.26300.9550]
(c) Microsoft Corporation. All rights reserved.
C:\Windows\System32>cd "C:\Program Files\SplunkUniversalForwarder\bin"
C:\Program Files\SplunkUniversalForwarder\bin>splunk add forward-server 192.168.56.101:9997
Your session is invalid. Please login.
Splunk username: michael1
Password:
192.168.56.101:9997 forwarded-server already present
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Active forwards:
192.168.56.101:9997
Configured but inactive forwards:
None
C:\Program Files\SplunkUniversalForwarder\bin>splunk add monitor "C:\Windows\System32\winevt\Logs\Application.evtx" -index windows
Added monitor of 'C:\Windows\System32\winevt\Logs\Application.evtx'.
C:\Program Files\SplunkUniversalForwarder\bin>splunk add monitor "C:\Windows\System32\winevt\Logs\Security.evtx" -index windows
Added monitor of 'C:\Windows\System32\winevt\Logs\Security.evtx'.
C:\Program Files\SplunkUniversalForwarder\bin>splunk add monitor "C:\Windows\System32\winevt\Logs\System.evtx" -index windows
Added monitor of 'C:\Windows\System32\winevt\Logs\System.evtx'.
C:\Program Files\SplunkUniversalForwarder\bin>
24°
7:33 am
08/10/2026

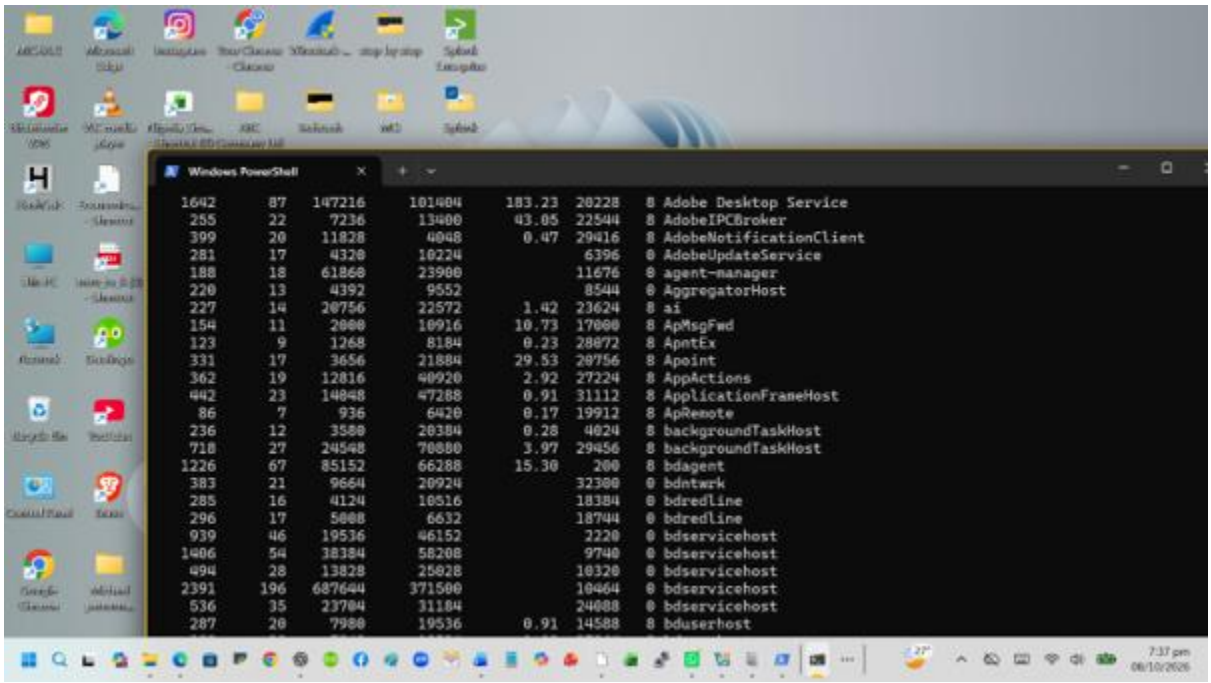

Administrator: Command Prompt
C:\Program Files\SplunkUniversalForwarder\bin>[WinEventLog://Application]disabled = 0index = windowsrenderXml = true[WinEventLog://System]disabled = 0index = windowsrenderXml = true[WinEventLog://Security]disabled = 0index = windowsrenderXml = true
The filename, directory name, or volume label syntax is incorrect.
C:\Program Files\SplunkUniversalForwarder\bin> WinEventLog://Application]disabled = 0index = windowsrenderXml = true WinEventLog://System disabled = 0index = windowsrenderXml = true WinEventLog://Security]disabled = 0index = windowsrenderXml = true
'WinEventLog:' is not recognized as an internal or external command,
operable program or batch file.
C:\Program Files\SplunkUniversalForwarder\bin>splunk restart
SplunkForwarder: Stopped
Splunk> All batbelt. No tights.
Checking prerequisites...
Checking mgmt port [8090]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from 'C:\Program Files\SplunkUniversalForwarder\splunkforwarder-10.6.0.5-86587d4e3b27-windows-x64-manifest'
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
SplunkForwarder: Starting (pid 2996)
Done
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
Splunk username: michael1
Password:
Login failed
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
25°
11:51 pm
05/10/2026

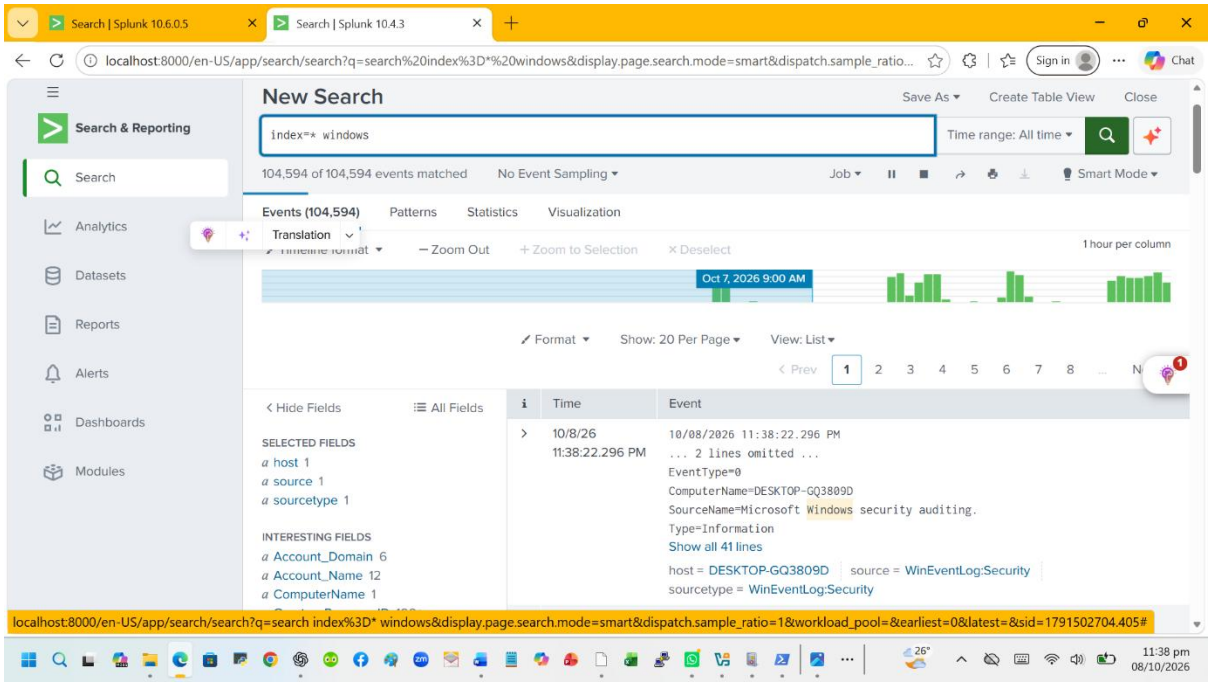

Administrator: Command Prompt
05/10/2026 09:49 pm <DIR> shcluster
05/10/2026 09:49 pm 559 splunk-launch.conf
05/10/2026 09:49 pm 559 splunk-launch.conf.default
28/09/2026 09:37 am 78 splunk.version
05/10/2026 09:49 pm <DIR> system
05/10/2026 09:49 pm <DIR> users
18 File(s) 65,707 bytes
13 Dir(s) 37,492,224,000 bytes free
C:\Program Files\SplunkUniversalForwarder\bin>dir "C:\Program Files\SplunkUniversalForwarder\etc\system\local"
Volume in drive C has no label.
Volume Serial Number is 2664-9909
Directory of C:\Program Files\SplunkUniversalForwarder\etc\system\local
05/10/2026 09:50 pm <DIR> .
05/10/2026 09:49 pm <DIR> ..
05/10/2026 09:49 pm 126 authentication.conf
05/10/2026 09:49 pm 72 deploymentclient.conf
05/10/2026 09:49 pm 156 outputs.conf
28/09/2026 09:11 am 273 README
05/10/2026 09:50 pm 508 server.conf
05/10/2026 09:49 pm 48 web.conf
6 File(s) 1,183 bytes
2 Dir(s) 37,461,159,936 bytes free
C:\Program Files\SplunkUniversalForwarder\bin>ren "C:\Program Files\SplunkUniversalForwarder\etc\passwd" passwd.old
C:\Program Files\SplunkUniversalForwarder\bin>notepad "C:\Program Files\SplunkUniversalForwarder\etc\system\local\user-seed.conf"
C:\Program Files\SplunkUniversalForwarder\bin>cd "C:\Program Files\SplunkUniversalForwarder\bin"
C:\Program Files\SplunkUniversalForwarder\bin>splunk start
Splunk> All batbelt. No tights.
Checking prerequisites...
Checking mgmt port [8090]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
Checking conf files for problems...
25°
11:50 pm
05/10/2026


Administrator: Command Prompt
Splunk> All batbelt. No tights.
Checking prerequisites...
Checking mgmt port [8090]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from 'C:\Program Files\SplunkUniversalForwarder\splunkforwarder-10.6.0.5-86587d4e3b27-windows-x64-manifest'
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
SplunkForwarder: Starting (pid 21872)
Done
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server -auth michael1:Michael@2026UF!
Active forwards:
192.168.56.101:9997
Configured but inactive forwards:
None
C:\Program Files\SplunkUniversalForwarder\bin>powershell Test-NetConnection 192.168.56.101 -Port 9997
ComputerName : 192.168.56.101
RemoteAddress : 192.168.56.101
RemotePort : 9997
InterfaceAlias : Ethernet 2
SourceAddress : 192.168.56.1
TcpTestSucceeded : True
C:\Program Files\SplunkUniversalForwarder\bin>cd "C:\Program Files\SplunkUniversalForwarder\bin"
11:50 pm
05/10/2026

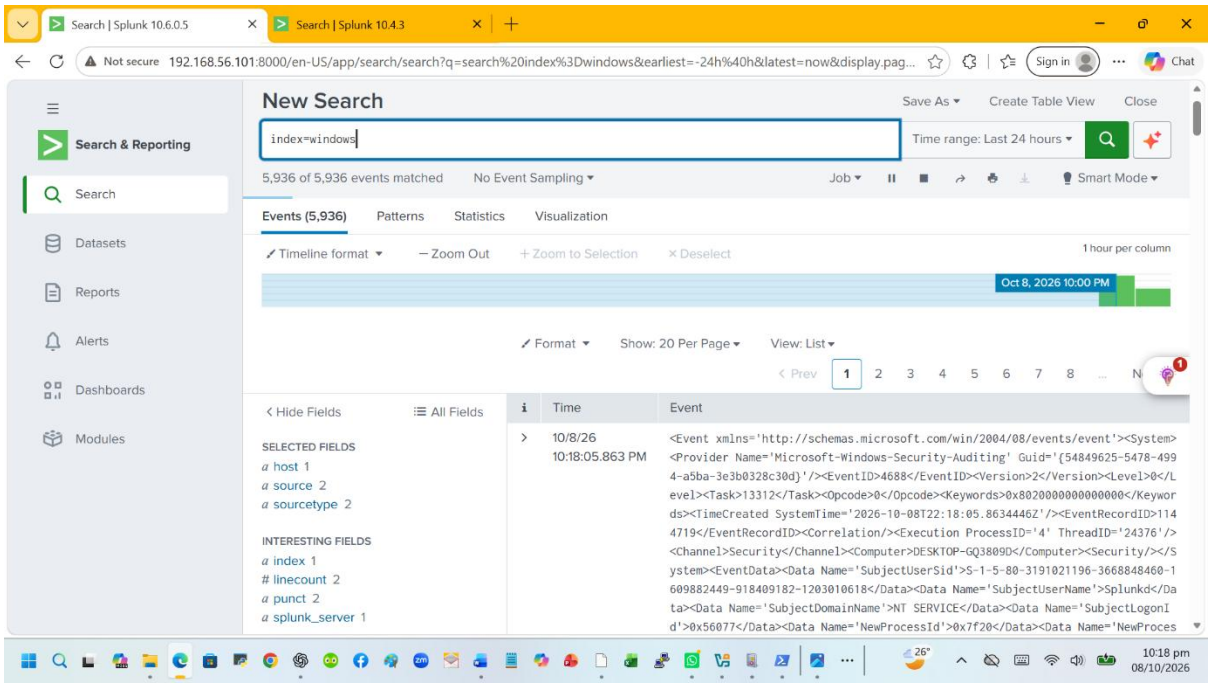

Administrator: Command Prompt
C:\Program Files\SplunkUniversalForwarder\bin>splunk stop
SplunkForwarder: Stopped
C:\Program Files\SplunkUniversalForwarder\bin>copy "C:\Program Files\SplunkUniversalForwarder\etc\passwd" "C:\Program Files\SplunkUniversalForwarder\etc\passwd.backup"
1 file(s) copied.
C:\Program Files\SplunkUniversalForwarder\bin>dir "C:\Program Files\SplunkUniversalForwarder\etc"
Volume in drive C has no label.
Volume Serial Number is 2664-9909
Directory of C:\Program Files\SplunkUniversalForwarder\etc
05/10/2026 10:54 pm <DIR> .
05/10/2026 09:49 pm <DIR> ..
05/10/2026 09:49 pm <DIR> apps
05/10/2026 09:50 pm <DIR> auth
28/09/2026 09:11 am 58 copyright.txt
28/09/2026 09:11 am 8,407 datetime.xml
05/10/2026 09:49 pm <DIR> deployment-apps
05/10/2026 09:49 pm <DIR> disabled-apps
05/10/2026 09:50 pm 56 instance.cfg
05/10/2026 09:49 pm <DIR> licenses
28/09/2026 09:14 am 514 log-btool-debug.cfg
28/09/2026 09:14 am 543 log-btool.cfg
28/09/2026 09:14 am 268 log-cmdline-debug.cfg
28/09/2026 09:14 am 2,731 log-cmdline.cfg
28/09/2026 09:14 am 7,116 log-debug.cfg
28/09/2026 09:14 am 7,017 log-node-platform.cfg
28/09/2026 09:14 am 514 log-tlsproxy.cfg
28/09/2026 09:14 am 1,380 log-utility.cfg
28/09/2026 09:14 am 33,685 log.cfg
05/10/2026 09:49 pm <DIR> manager-apps
05/10/2026 09:49 pm <DIR> modules
05/10/2026 09:49 pm <DIR> myinstall
05/10/2026 09:49 pm 166 passwd
05/10/2026 09:49 pm 166 passwd.backup
28/09/2026 09:11 am 1,890 prettyprint.xsl
05/10/2026 09:49 pm <DIR> shcluster
05/10/2026 09:49 pm 559 splunk-launch.conf
05/10/2026 09:49 pm 559 splunk-launch.conf.default
11:50 pm
05/10/2026


Administrator: Command Prompt
Splunk> All batbelt. No tights.
Checking prerequisites...
Checking mgmt port [8090]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from 'C:\Program Files\SplunkUniversalForwarder\splunkforwarder-10.6.0.5-86587d4e3b27-windows-x64-manifest'
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
SplunkForwarder: Starting (pid 21872)
Done
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server -auth michael1:Michael@2026UF!
Active forwards:
192.168.56.101:9997
Configured but inactive forwards:
None
C:\Program Files\SplunkUniversalForwarder\bin>powershell Test-NetConnection 192.168.56.101 -Port 9997
ComputerName : 192.168.56.101
RemoteAddress : 192.168.56.101
RemotePort : 9997
InterfaceAlias : Ethernet 2
SourceAddress : 192.168.56.1
TcpTestSucceeded : True
C:\Program Files\SplunkUniversalForwarder\bin>
11:09 pm
05/10/2026


Windows PowerShell
1642 87 147216 101404 183.23 20228 8 Adobe Desktop Service
255 22 7236 13400 43.05 22544 8 AdobeIPCBroker
399 20 11828 4048 0.47 29416 8 AdobeNotificationClient
281 17 4320 10224 6396 0 AdobeUpdateService
188 18 61868 23900 11676 0 agent-manager
220 13 4392 9552 8544 0 AggregatorHost
227 14 20756 22572 1.42 23624 8 ai
154 11 2008 10916 10.73 17080 8 ApMsgFwd
123 9 1268 8184 0.23 28072 8 ApntEx
331 17 3656 21884 29.53 20756 8 Apoint
362 19 12816 40920 2.92 27224 8 AppActions
442 23 14048 47288 0.91 31112 8 ApplicationFrameHost
86 7 936 6420 0.17 19912 8 ApRemote
236 12 3580 20384 0.28 4024 8 backgroundTaskHost
718 27 24548 70880 3.97 29456 8 backgroundTaskHost
1226 67 85152 66288 15.30 200 8 bdagent
383 21 9664 20924 32300 0 bdntwrk
285 16 4124 10516 18384 0 bdredline
296 17 5088 6632 18744 0 bdredline
939 46 19536 46152 2220 0 bdservicehost
1406 54 38384 58208 9740 0 bdservicehost
494 28 13828 25028 10320 0 bdservicehost
2391 196 687644 371500 10464 0 bdservicehost
536 35 23704 31184 24088 0 bdservicehost
287 20 7980 19536 0.91 14588 8 bduserhost
7:37 pm
06/10/2026



This is the screenshot of window logs ingestion. This is to show that our window endpoint is connected to our SIEM Splunk



6. Linux Logs Collection

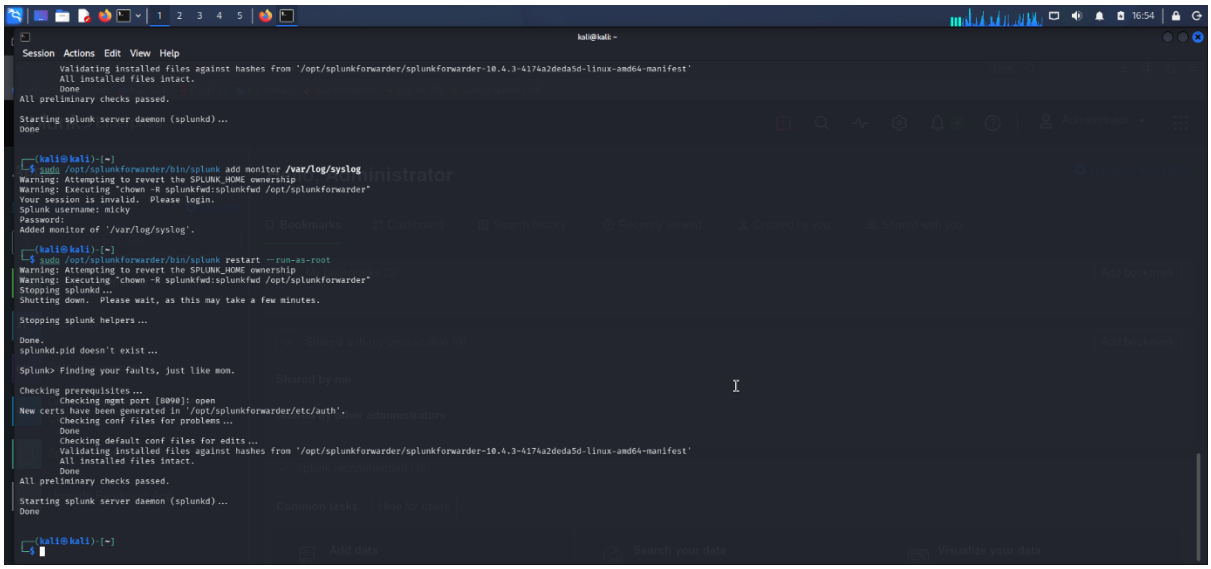
The Linux endpoint was also connected to the Splunk server using a Splunk Universal Forwarder.

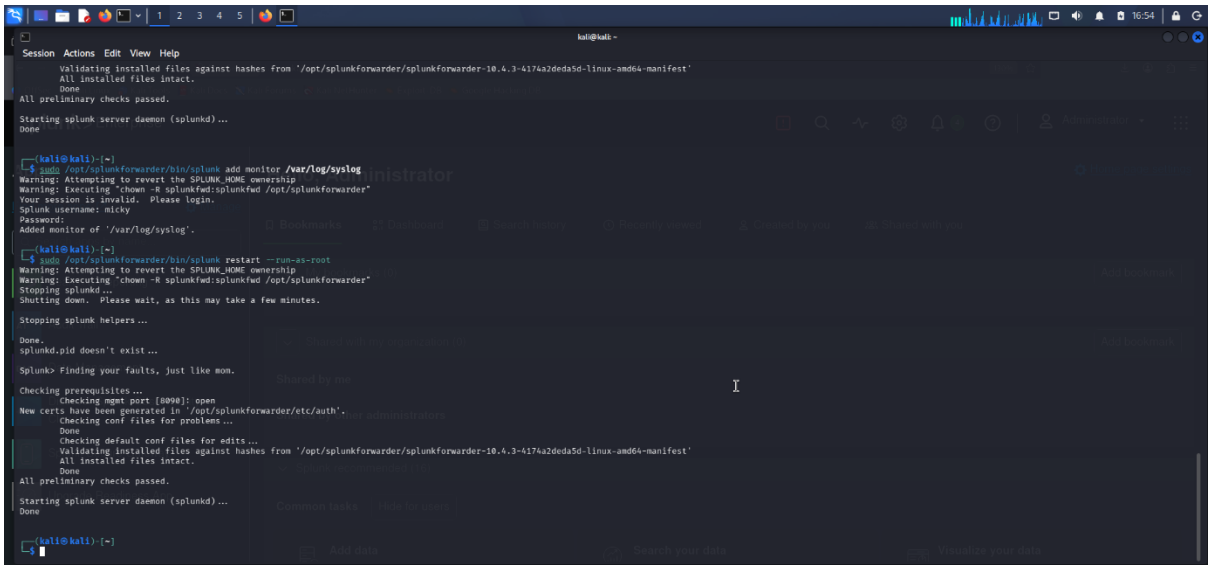
Important Linux logs included:

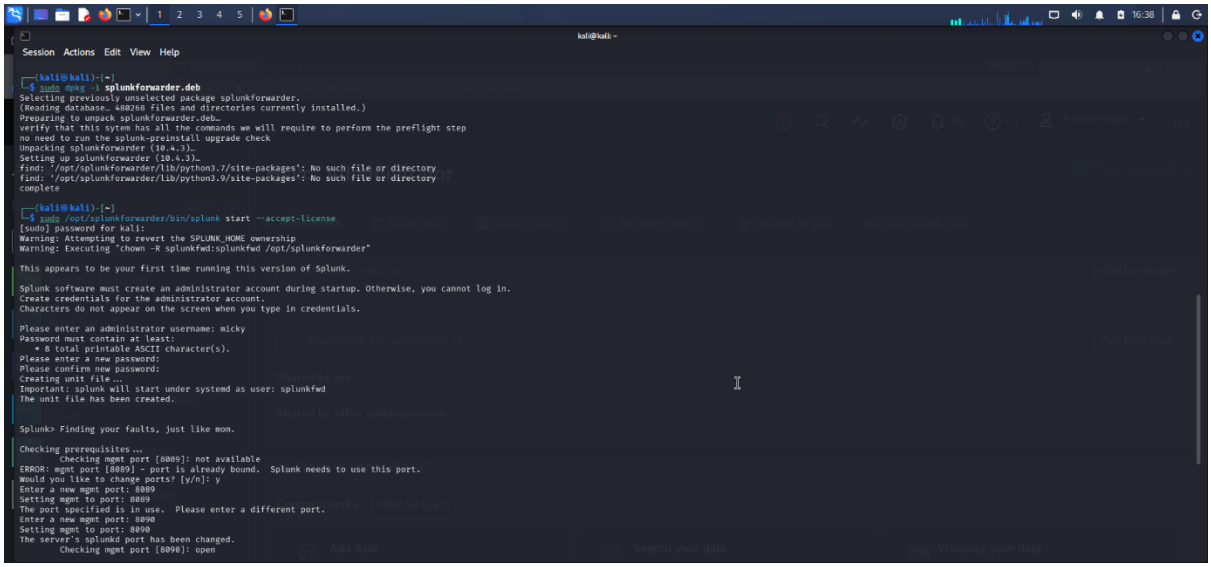
- /var/log/auth.log
- /var/log/syslog

The authentication log was particularly useful for identifying failed login attempts, successful authentication, privilege-related activity, and other security events.

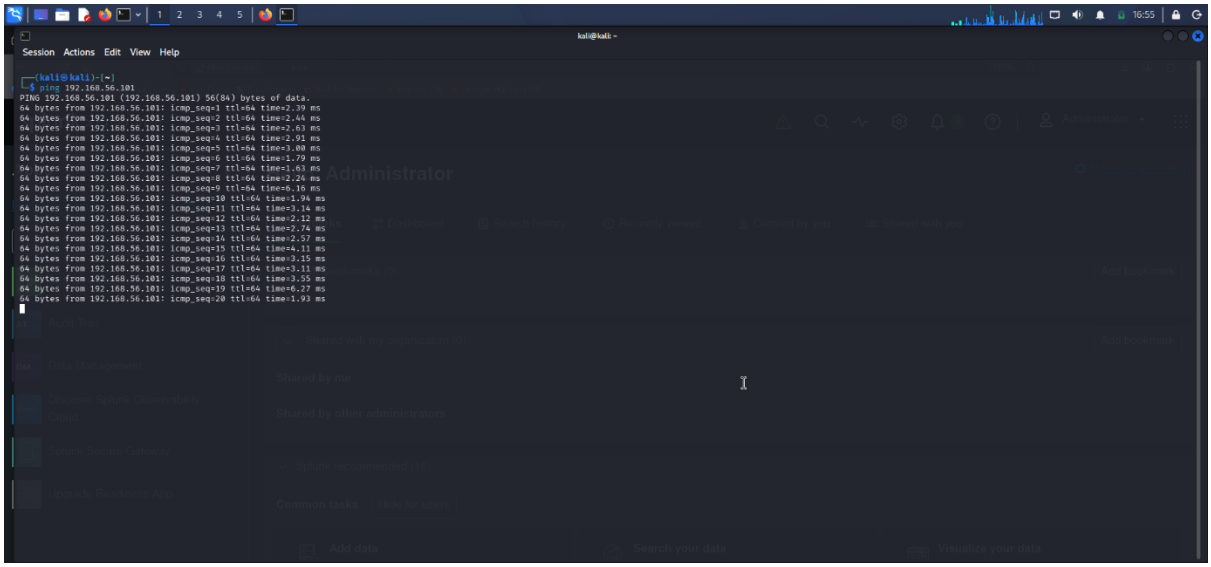
Please, kindly find the attached screenshot below as evidence:

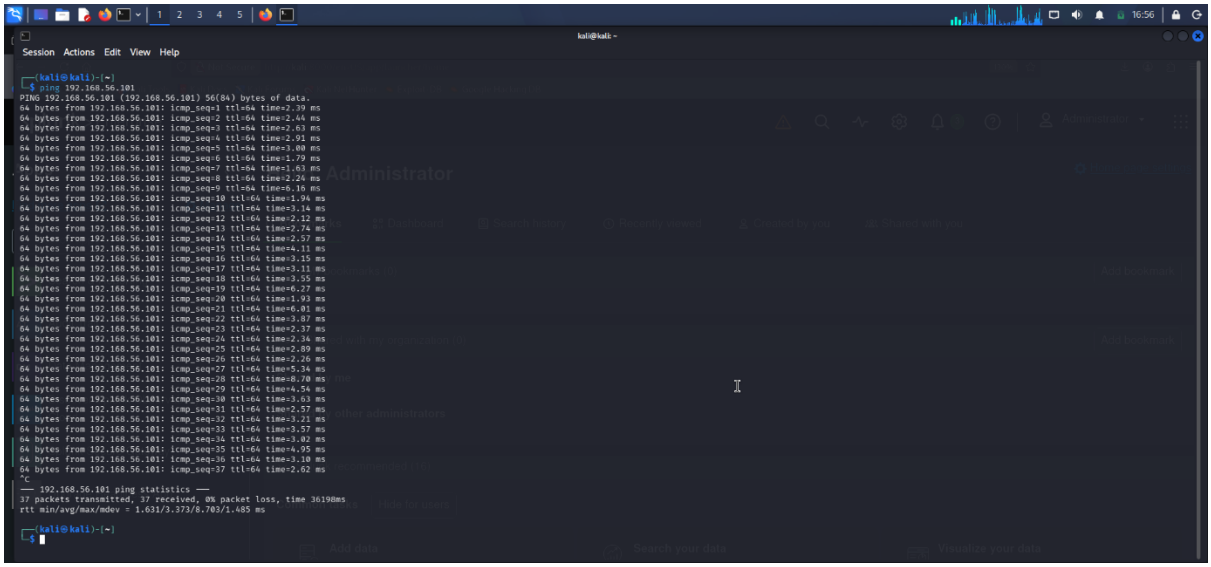




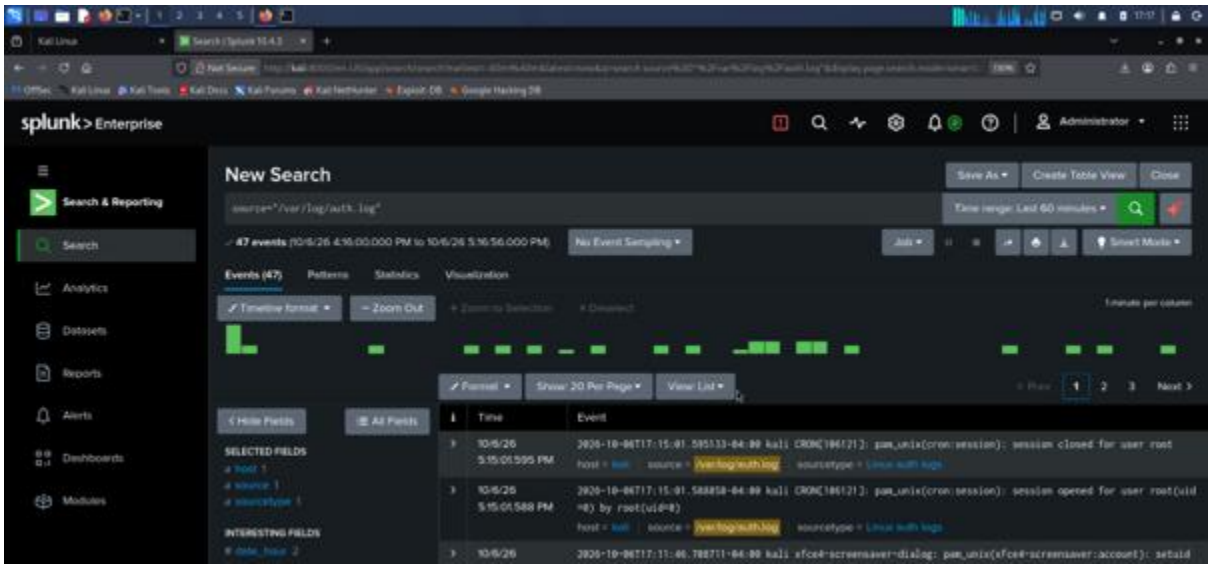

kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo dpkg -i splunkforwarder.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 440268 files and directories currently installed.)
Preparing to unpack splunkforwarder.deb ...
verify that this sytem has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunkforwarder (10.4.3) ...
Setting up splunkforwarder (10.4.3) ...
find: '/opt/splunkforwarder/lib/python3.7/site-packages': No such file or directory
find: '/opt/splunkforwarder/lib/python3.9/site-packages': No such file or directory
complete
(kali㉿kali)-[~]
$ sudo /opt/splunkforwarder/bin/splunk start --accept-license
[sudo] password for kali:
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
This appears to be your first time running this version of Splunk.
Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.
Please enter an administrator username: micky
Password must contain at least:
• 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file ...
Important: splunk will start under systemd as user: splunkfwd
The unit file has been created.
Splunk> Finding your faults, just like mom.
Checking prerequisites ...
Checking mgmt port [8089]: not available
ERROR: mgmt port [8089] - port is already bound. Splunk needs to use this port.
Would you like to change ports? [y/n]: y
Enter a new mgmt port: 8089
Setting mgmt to port: 8089
The port specified is in use. Please enter a different port.
Enter a new mgmt port: 8090
Setting mgmt to port: 8090
The server's splunkd port has been changed.
Checking mgmt port [8090]: open

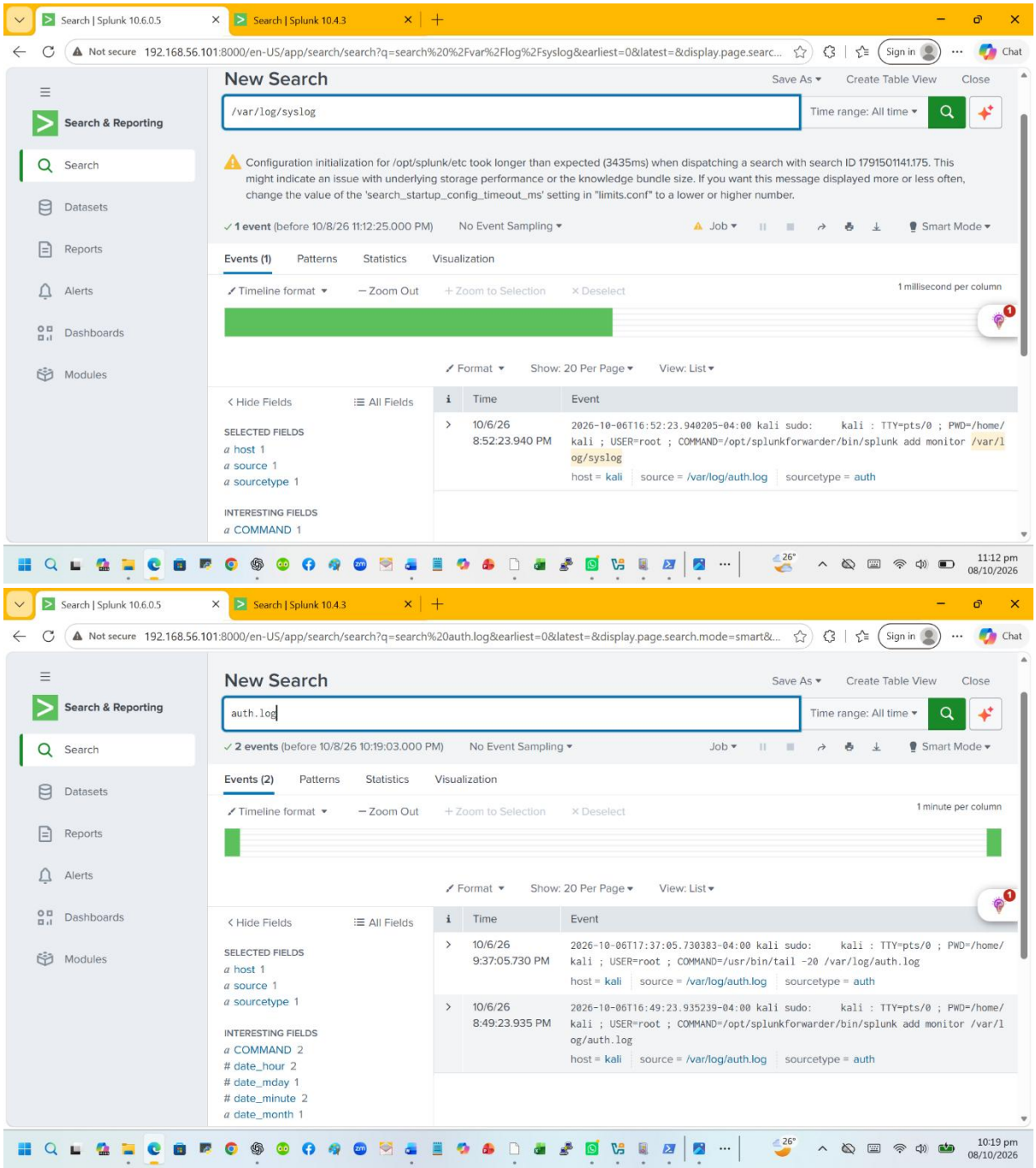



kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=2.39 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=2.44 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.63 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=2.91 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=3.00 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=1.79 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=1.63 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=2.24 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=6.16 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=1.94 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=3.14 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=2.12 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=2.74 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=2.57 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=4.11 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=3.15 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=3.11 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=3.55 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=6.27 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=1.93 ms


Session Actions Edit View Help
kali@kali ~
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=2.39 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=2.44 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.63 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=2.91 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.09 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=1.79 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=1.63 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=2.24 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=6.16 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=1.94 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=3.14 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=2.12 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=2.74 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=2.57 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=4.11 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=3.15 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=3.11 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=3.55 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=6.27 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=1.93 ms
64 bytes from 192.168.56.101: icmp_seq=21 ttl=64 time=6.01 ms
64 bytes from 192.168.56.101: icmp_seq=22 ttl=64 time=3.87 ms
64 bytes from 192.168.56.101: icmp_seq=23 ttl=64 time=2.37 ms
64 bytes from 192.168.56.101: icmp_seq=24 ttl=64 time=2.34 ms
64 bytes from 192.168.56.101: icmp_seq=25 ttl=64 time=2.89 ms
64 bytes from 192.168.56.101: icmp_seq=26 ttl=64 time=2.26 ms
64 bytes from 192.168.56.101: icmp_seq=27 ttl=64 time=5.34 ms
64 bytes from 192.168.56.101: icmp_seq=28 ttl=64 time=8.70 ms
64 bytes from 192.168.56.101: icmp_seq=29 ttl=64 time=4.54 ms
64 bytes from 192.168.56.101: icmp_seq=30 ttl=64 time=3.63 ms
64 bytes from 192.168.56.101: icmp_seq=31 ttl=64 time=2.57 ms
64 bytes from 192.168.56.101: icmp_seq=32 ttl=64 time=3.21 ms
64 bytes from 192.168.56.101: icmp_seq=33 ttl=64 time=3.57 ms
64 bytes from 192.168.56.101: icmp_seq=34 ttl=64 time=3.02 ms
64 bytes from 192.168.56.101: icmp_seq=35 ttl=64 time=4.95 ms
64 bytes from 192.168.56.101: icmp_seq=36 ttl=64 time=3.10 ms
64 bytes from 192.168.56.101: icmp_seq=37 ttl=64 time=2.62 ms
^C
— 192.168.56.101 ping statistics —
37 packets transmitted, 37 received, 0% packet loss, time 36198ms
rtt min/avg/max/mdev = 1.631/3.373/8.703/1.485 ms
kali@kali ~


Session Actions Edit View Help


splunk> Enterprise
Administrator
Search & Reporting
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Modules
New Search
Save As
Create Table View
Close
source="/var/log/auth.log"
Time range: Last 60 minutes
47 events (10/6/26 4:16:00.000 PM to 10/6/26 5:16:56.000 PM)
No Event Sampling
Job
Smart Mode
Events (47)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Format
Show: 20 Per Page
View: List
Prev
1
2
3
Next
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 1
sourcetype 1
INTERESTING FIELDS
Time
Event
10/6/26 5:15:01.595 PM
2026-10-06T17:15:01.595133-04:00 kali CRON[106121]: pam_unix(cron:session): session closed for user root
10/6/26 5:15:01.588 PM
2026-10-06T17:15:01.588858-04:00 kali CRON[106121]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
host = kali
source = /var/log/auth.log
sourcetype = Linux auth logs
10/6/26
2026-10-06T17:11:46.788711-04:00 kali xfce4-screensaver-dialog: pam_unix(xfce4-screensaver:account): acsuid



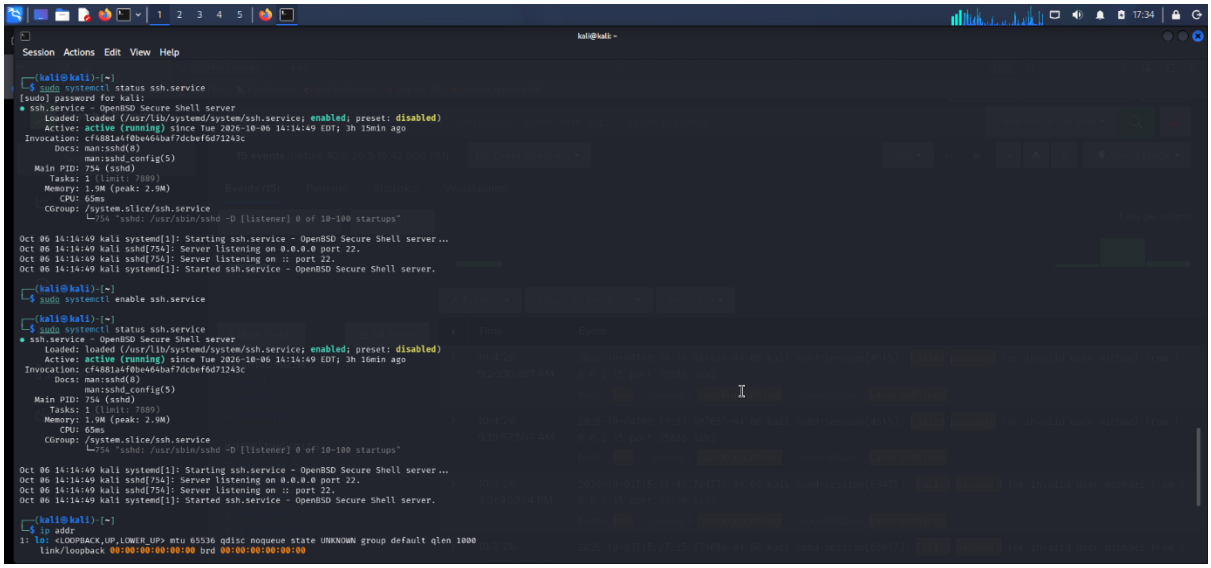
7. SIMULATED SECURITY ATTACKS

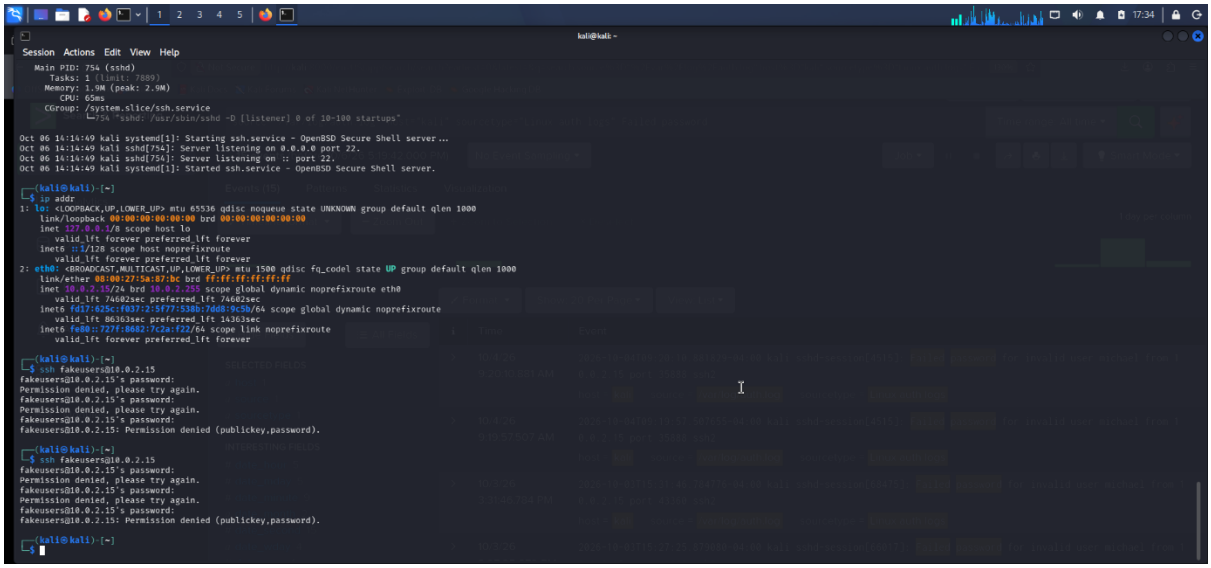
Four controlled attack scenarios were performed to test the ability of the SIEM to detect suspicious activities and to check the counter response from our SOC Analysts

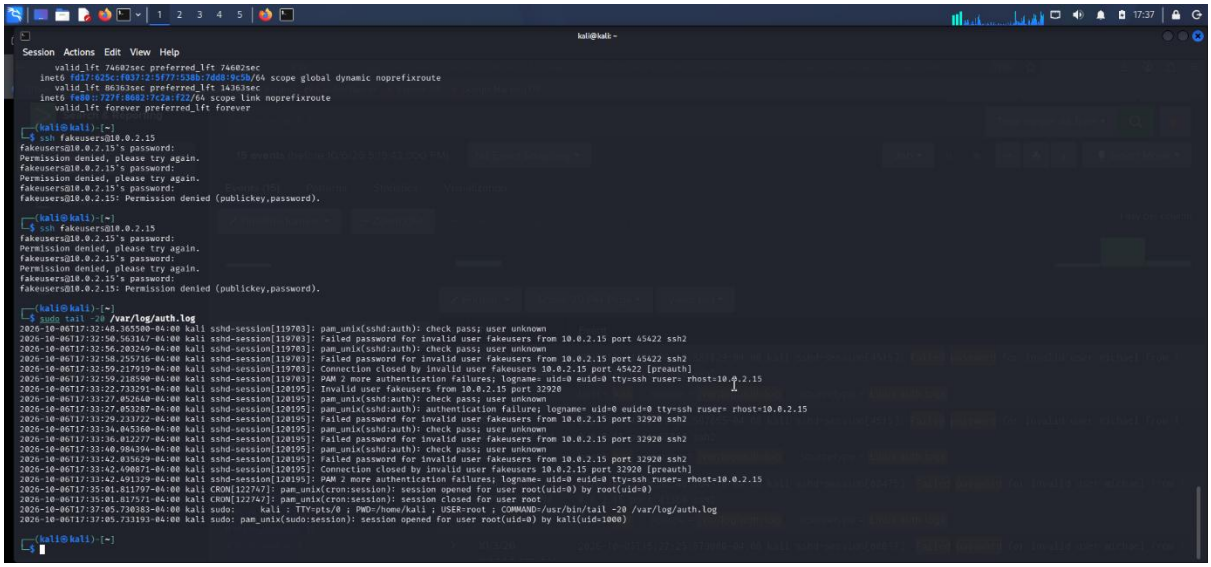
7.1 Brute-Force Login Attempt

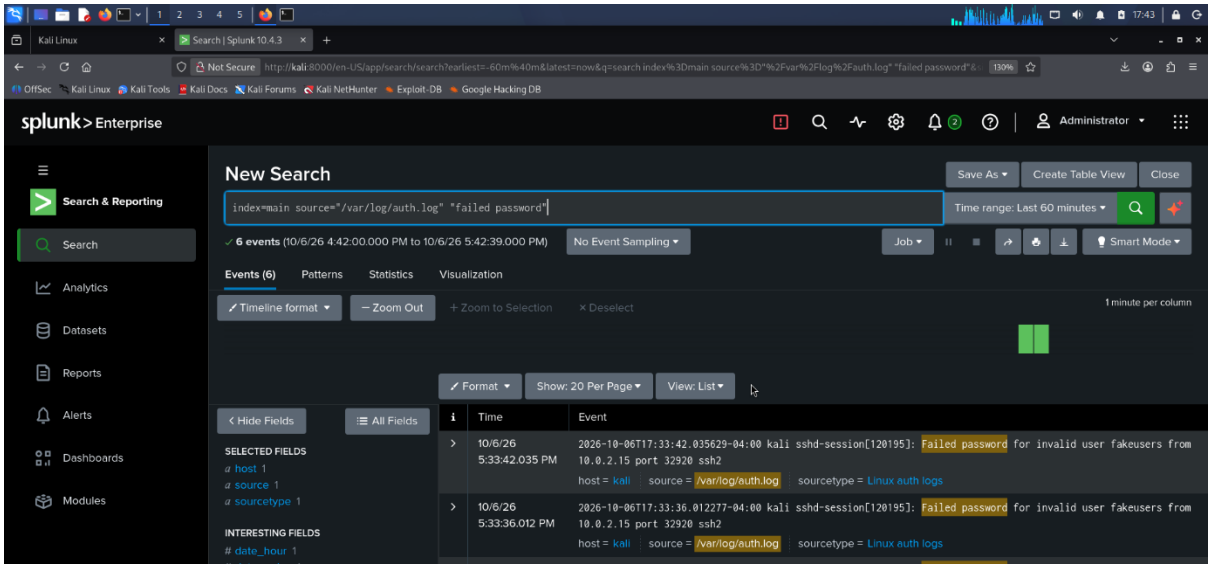
Multiple unsuccessful authentication attempts were generated against the designated Linux test system. The login attempt was tried six times in two minutes.

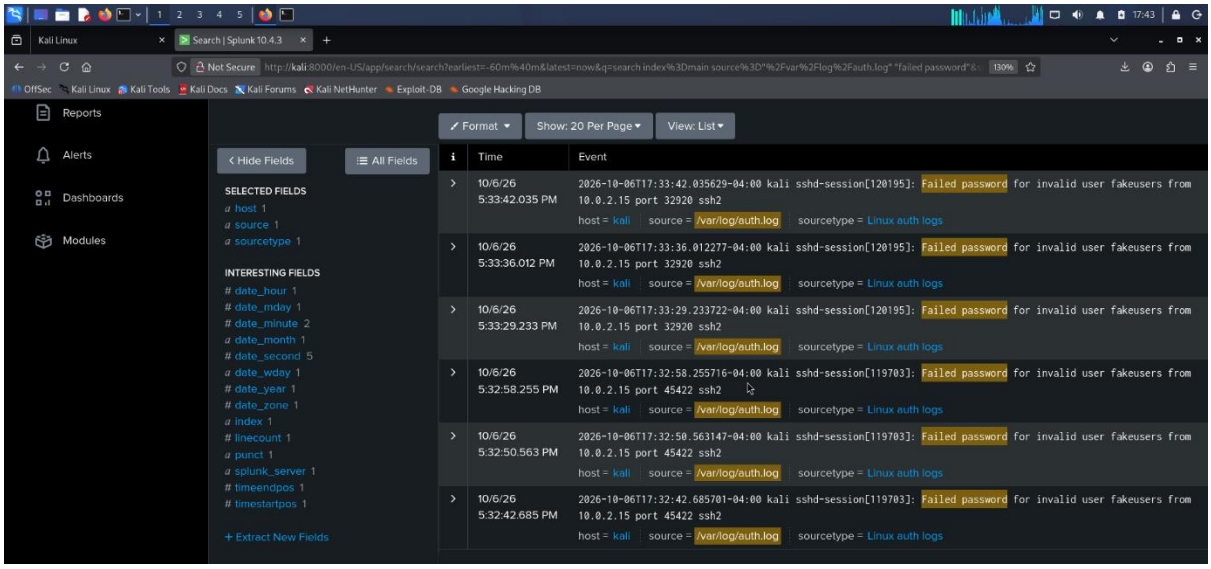
An account called **fakeusers** was seen making an attempt to make brute force attempt to break into company system. He or she tried it 6 times in a roll.






kali@kali ~
Session Actions Edit View Help
ssh fakeusers@10.0.2.15
fakeusers@10.0.2.15's password:
Permission denied, please try again.
fakeusers@10.0.2.15's password:
Permission denied, please try again.
fakeusers@10.0.2.15's password:
fakeusers@10.0.2.15: Permission denied (publickey,password).
sudo tail -20 /var/log/auth.log
2026-10-06T17:32:48.365500-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:50.563147-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:56.203249-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:58.255716-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:59.217919-04:00 kali sshd-session[119703]: Connection closed by invalid user fakeusers 10.0.2.15 port 45422 [preauth]
2026-10-06T17:32:59.218590-04:00 kali sshd-session[119703]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:33:22.733291-04:00 kali sshd-session[120195]: Invalid user fakeusers from 10.0.2.15 port 32920
2026-10-06T17:33:27.052640-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:27.053287-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:33:29.233722-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:34.045360-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:36.012277-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:40.984394-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:42.035629-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:42.690871-04:00 kali sshd-session[120195]: Connection closed by invalid user fakeusers 10.0.2.15 port 32920 [preauth]
2026-10-06T17:33:42.491329-04:00 kali sshd-session[120195]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:35:01.811797-04:00 kali CRON[122747]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-10-06T17:35:01.817571-04:00 kali CRON[122747]: pam_unix(cron:session): session closed for user root
2026-10-06T17:37:05.730383-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -20 /var/log/auth.log
2026-10-06T17:37:05.733193-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)


splunk> Enterprise
Administrator
Search & Reporting
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Modules
New Search
Save As
Create Table View
Close
index=main source="/var/log/auth.log" "failed password"
Time range: Last 60 minutes
6 events (10/6/26 4:42:00.000 PM to 10/6/26 5:42:39.000 PM)
No Event Sampling
Job
Smart Mode
Events (6)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 minute per column
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 1
sourcetype 1
INTERESTING FIELDS
date_hour 1
Time
Event
10/6/26 5:33:42.035 PM
2026-10-06T17:33:42.035629-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs
10/6/26 5:33:36.012 PM
2026-10-06T17:33:36.012277-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs


Reports
Alerts
Dashboards
Modules
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 1
sourcetype 1
INTERESTING FIELDS
date_hour 1
date_mday 1
date_minute 2
date_month 1
date_second 5
date_wday 1
date_year 1
date_zone 1
index 1
linecount 1
punct 1
splunk_server 1
timeendpos 1
timestartpos 1
Extract New Fields
Time
Event
10/6/26 5:33:42.035 PM
2026-10-06T17:33:42.035629-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs
10/6/26 5:33:36.012 PM
2026-10-06T17:33:36.012277-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs
10/6/26 5:33:29.233 PM
2026-10-06T17:33:29.233722-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs
10/6/26 5:32:58.255 PM
2026-10-06T17:32:58.255716-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs
10/6/26 5:32:50.563 PM
2026-10-06T17:32:50.563147-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs
10/6/26 5:32:42.685 PM
2026-10-06T17:32:42.685701-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
host = kali source = /var/log/auth.log sourcetype = Linux auth logs

The resulting authentication failures were collected by Splunk. Repeated failures from the same source were investigated as potential brute-force activity.

A Splunk search like the following was used:

index=* **"Failed password" or Authenticate failure”**

The investigation focused on the source IP address, username, destination host, number of attempts, and timestamp.

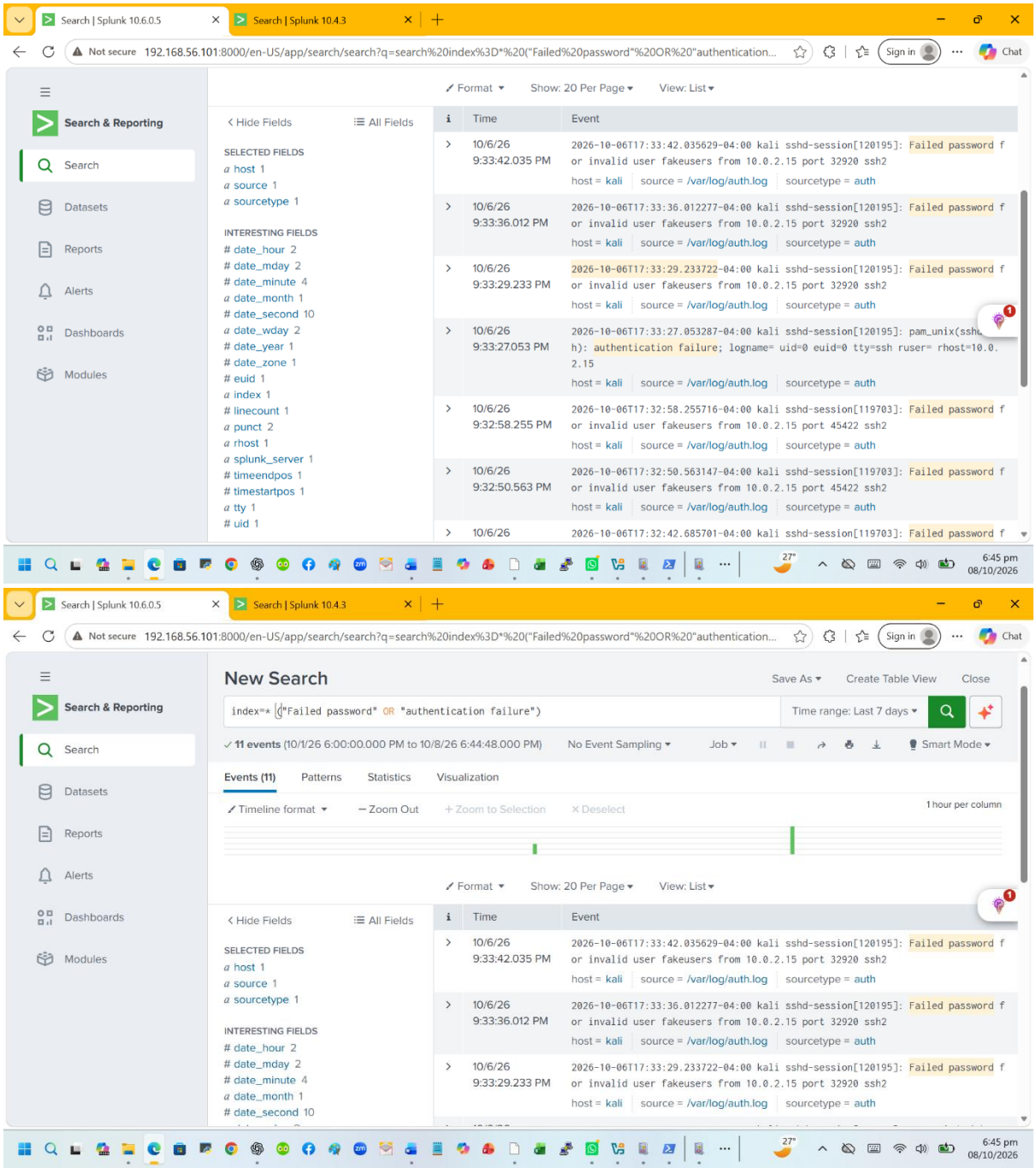
Attacker’s account- fakeusers

Affected host – Kali Linux

Date of the Attacks -06/10/2026

Time of attacks - 5:33PM

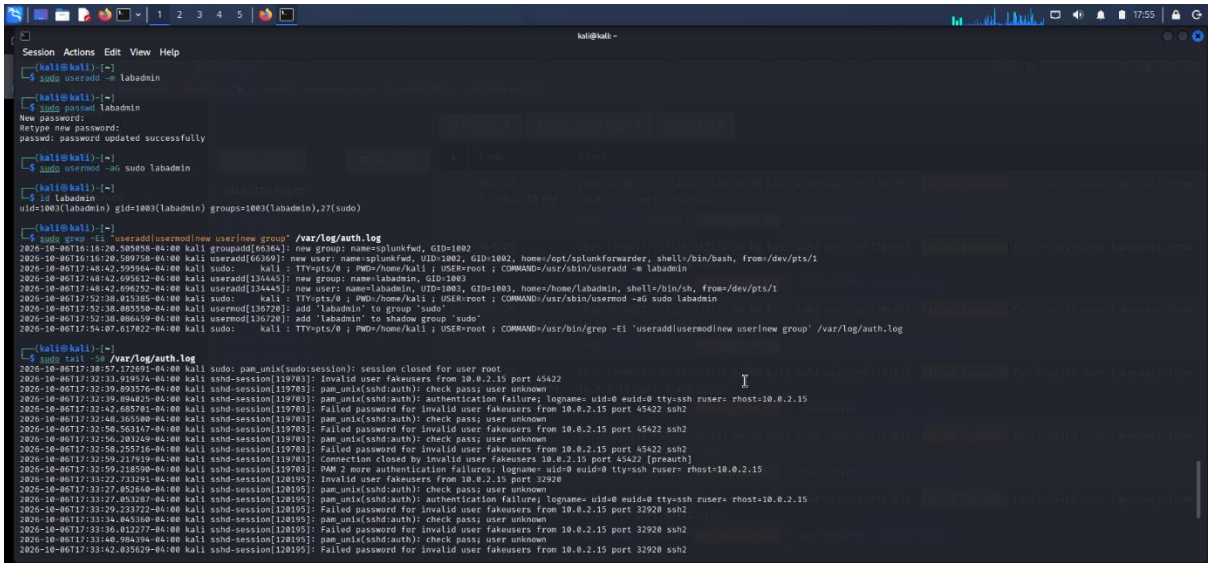
MITRE ATT&CK: T1110 – Brute Force.

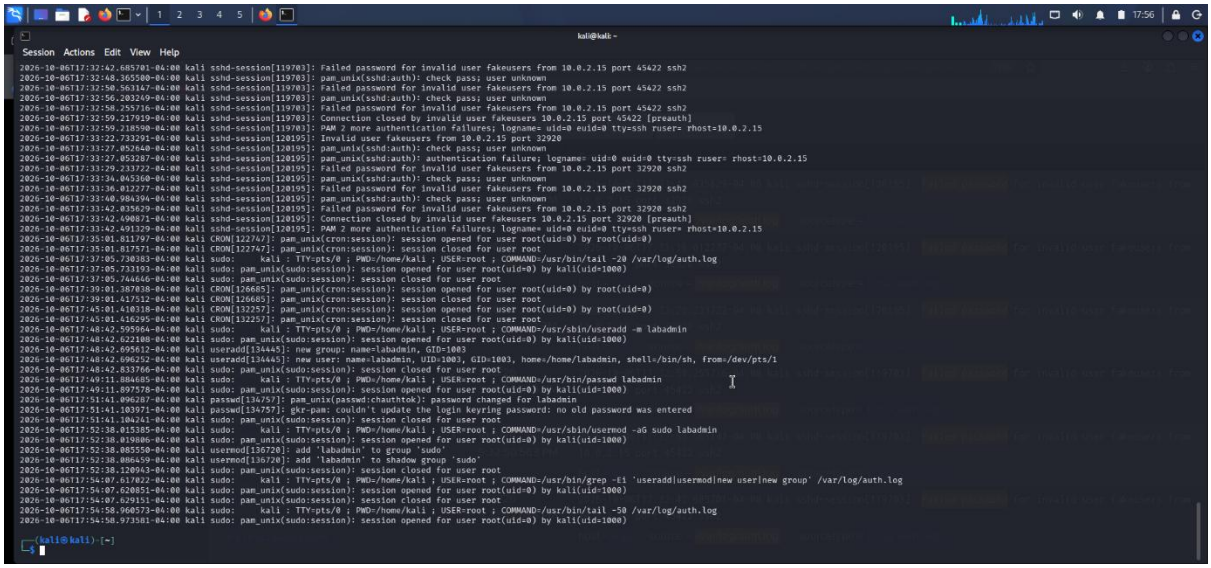


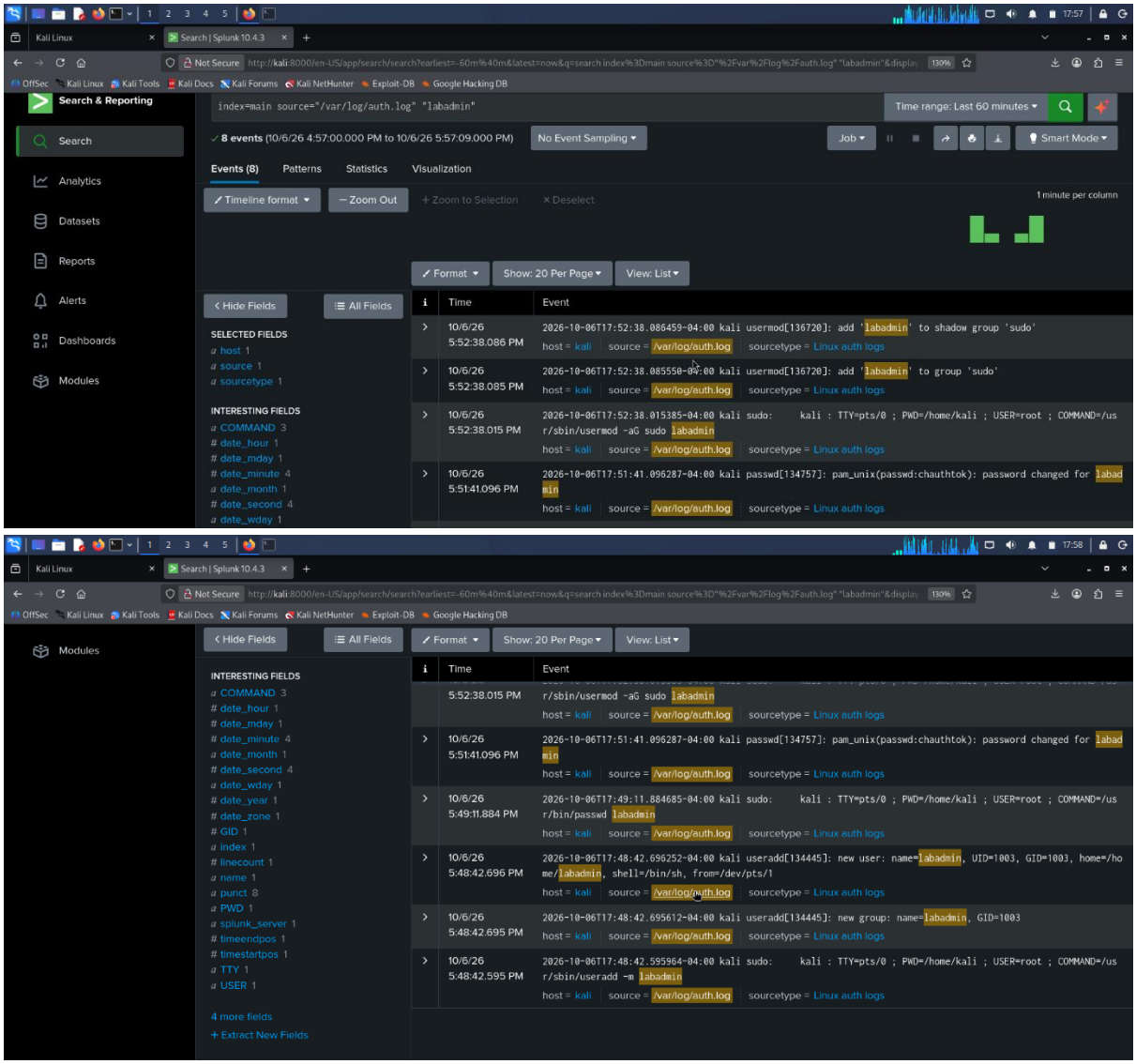
7.2 Unauthorized Account Creation

A test account was created on the designated endpoint (Linux) to simulate an attacker creating an additional account, in order to get access to admin system.

An account name **labadmin** was created in order to get unauthorised access.


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo useradd -m labadmin
(kali㉿kali)-[~]
$ sudo passwd labadmin
New password:
Retype new password:
passwd: password updated successfully
(kali㉿kali)-[~]
$ sudo usermod -aG sudo labadmin
(kali㉿kali)-[~]
$ id labadmin
uid=1003(labadmin) gid=1003(labadmin) groups=1003(labadmin),27(sudo)
(kali㉿kali)-[~]
$ sudo grep -Ei "useradd|usermod|new user|new group" /var/log/auth.log
2026-10-06T16:16:20.505058-04:00 kali groupadd[66364]: new group: name=splunkfwd, GID=1002
2026-10-06T16:16:20.509758-04:00 kali useradd[66366]: new user: name=splunkfwd, UID=1002, GID=1002, home=/opt/splunkforwarder, shell=/bin/bash, from=/dev/pts/1
2026-10-06T17:48:42.595964-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/useradd -m labadmin
2026-10-06T17:48:42.695612-04:00 kali useradd[134445]: new group: name=labadmin, GID=1003
2026-10-06T17:48:42.696252-04:00 kali useradd[134445]: new user: name=labadmin, UID=1003, GID=1003, home=/home/labadmin, shell=/bin/sh, from=/dev/pts/1
2026-10-06T17:52:38.015385-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/usermod -aG sudo labadmin
2026-10-06T17:52:38.085550-04:00 kali usermod[136720]: add 'labadmin' to group 'sudo'
2026-10-06T17:52:38.086459-04:00 kali usermod[136720]: add 'labadmin' to shadow group 'sudo'
2026-10-06T17:54:07.617022-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/grep -Ei 'useradd|usermod|new user|new group' /var/log/auth.log
(kali㉿kali)-[~]
$ sudo tail -50 /var/log/auth.log
2026-10-06T17:30:57.172691-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-06T17:32:33.919574-04:00 kali sshd-session[119703]: Invalid user fakeusers from 10.0.2.15 port 45422
2026-10-06T17:32:39.893576-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:39.894025-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:32:42.685701-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:48.365500-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:50.563147-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:56.203249-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:58.255716-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:59.217919-04:00 kali sshd-session[119703]: Connection closed by invalid user fakeusers 10.0.2.15 port 45422 [preauth]
2026-10-06T17:32:59.218590-04:00 kali sshd-session[119703]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:33:22.733291-04:00 kali sshd-session[120195]: Invalid user fakeusers from 10.0.2.15 port 32920
2026-10-06T17:33:27.052640-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:27.053287-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:33:29.233722-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:34.043360-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:36.012277-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:40.984394-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:42.035629-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2


kali@kali: ~
Session Actions Edit View Help
2026-10-06T17:32:42.685701-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:48.365500-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:50.563147-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:56.203249-04:00 kali sshd-session[119703]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:32:58.255716-04:00 kali sshd-session[119703]: Failed password for invalid user fakeusers from 10.0.2.15 port 45422 ssh2
2026-10-06T17:32:59.217919-04:00 kali sshd-session[119703]: Connection closed by invalid user fakeusers 10.0.2.15 port 45422 [preauth]
2026-10-06T17:32:59.218590-04:00 kali sshd-session[119703]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:33:22.733291-04:00 kali sshd-session[120195]: Invalid user fakeusers from 10.0.2.15 port 32920
2026-10-06T17:33:27.052640-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:27.053287-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:33:29.233722-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:34.043360-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:36.012277-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:40.984394-04:00 kali sshd-session[120195]: pam_unix(sshd:auth): check pass; user unknown
2026-10-06T17:33:42.035629-04:00 kali sshd-session[120195]: Failed password for invalid user fakeusers from 10.0.2.15 port 32920 ssh2
2026-10-06T17:33:42.490871-04:00 kali sshd-session[120195]: Connection closed by invalid user fakeusers 10.0.2.15 port 32920 [preauth]
2026-10-06T17:33:42.491329-04:00 kali sshd-session[120195]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15
2026-10-06T17:35:01.811797-04:00 kali CRON[122747]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-10-06T17:35:01.817571-04:00 kali CRON[122747]: pam_unix(cron:session): session closed for user root
2026-10-06T17:37:05.730383-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -20 /var/log/auth.log
2026-10-06T17:37:05.733193-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
2026-10-06T17:37:05.744464-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-06T17:39:01.387038-04:00 kali CRON[126685]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-10-06T17:39:01.417512-04:00 kali CRON[126685]: pam_unix(cron:session): session closed for user root
2026-10-06T17:45:01.410316-04:00 kali CRON[132257]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-10-06T17:45:01.416295-04:00 kali CRON[132257]: pam_unix(cron:session): session closed for user root
2026-10-06T17:48:42.595964-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/useradd -m labadmin
2026-10-06T17:48:42.622108-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
2026-10-06T17:48:42.695612-04:00 kali useradd[134445]: new group: name=labadmin, GID=1003
2026-10-06T17:48:42.696252-04:00 kali useradd[134445]: new user: name=labadmin, UID=1003, GID=1003, home=/home/labadmin, shell=/bin/sh, from=/dev/pts/1
2026-10-06T17:48:42.833766-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-06T17:49:11.804605-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/passwd labadmin
2026-10-06T17:49:11.897578-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
2026-10-06T17:51:41.096287-04:00 kali passwd[134757]: pam_unix(passwd:chauthtok): password changed for labadmin
2026-10-06T17:51:41.103971-04:00 kali passwd[134757]: gkr-pam: couldn't update the login keyring password: no old password was entered
2026-10-06T17:51:41.104241-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-06T17:52:38.015385-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/usermod -aG sudo labadmin
2026-10-06T17:52:38.019806-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
2026-10-06T17:52:38.085550-04:00 kali usermod[136720]: add 'labadmin' to group 'sudo'
2026-10-06T17:52:38.086459-04:00 kali usermod[136720]: add 'labadmin' to shadow group 'sudo'
2026-10-06T17:52:38.120943-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-06T17:54:07.617022-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/grep -Ei 'useradd|usermod|new user|new group' /var/log/auth.log
2026-10-06T17:54:07.620851-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
2026-10-06T17:54:07.629151-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-06T17:54:58.960573-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -50 /var/log/auth.log
2026-10-06T17:54:58.973581-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
(kali㉿kali)-[~]
$



The investigation identified the newly created account, the account that performed the action, the affected host, and the time of the event.

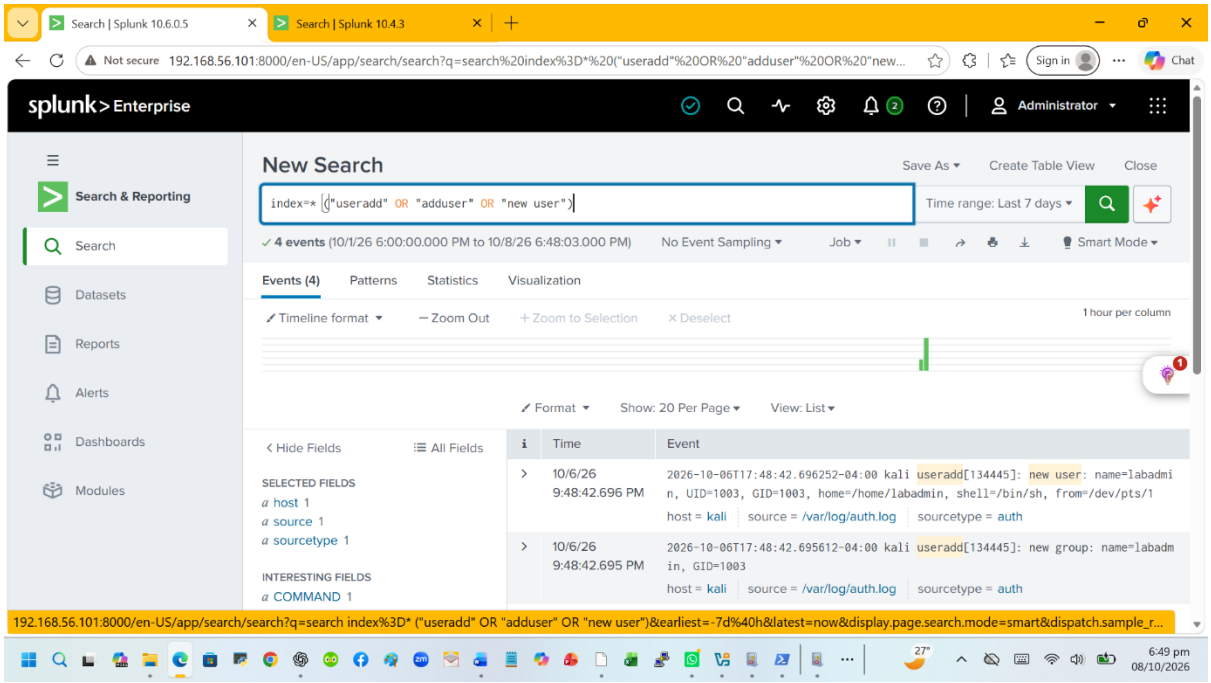
Attacker's account-labadmin

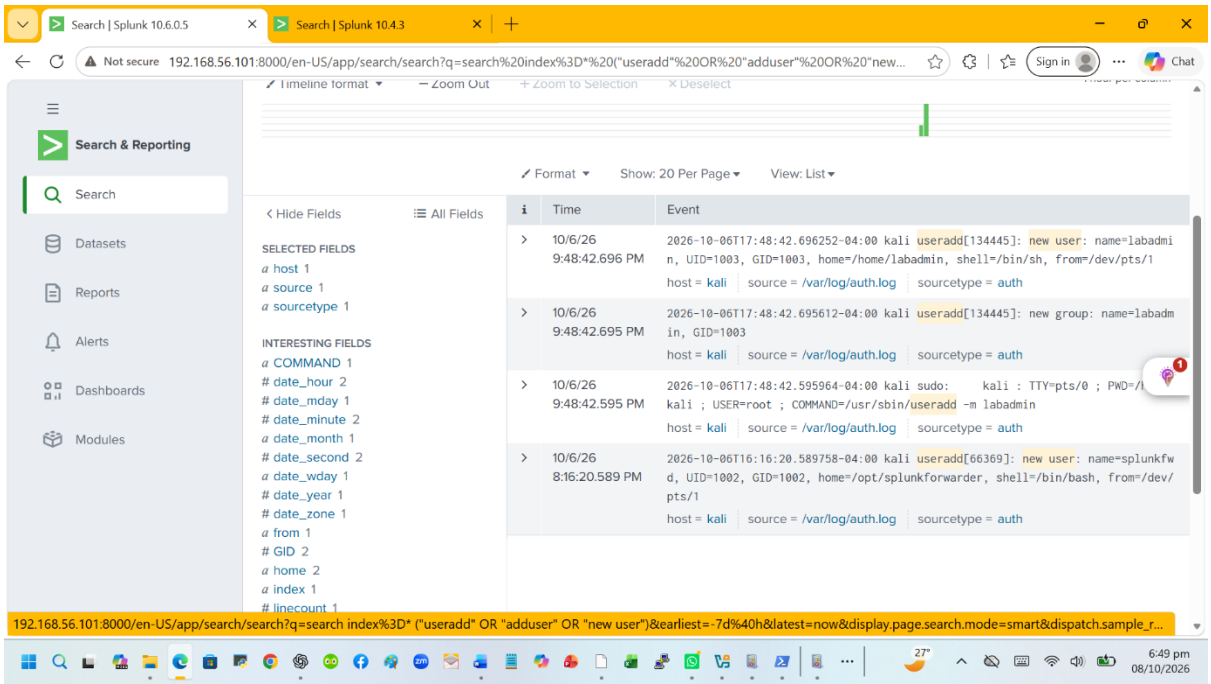
Affected host – Kali Linux

Date of the Attacks -06/10/2026

Time of attacks -5:48PM

MITRE ATT&CK: T1136 – Create Account


Search | Splunk 10.6.0.5
Search | Splunk 10.4.3
Not secure 192.168.56.101:8000/en-US/app/search/search?q=search%20index%3D*%20("useradd"%20OR%20"adduser"%20OR%20"new...
Sign in
Chat
splunk> Enterprise
Administrator
Search & Reporting
Search
Datasets
Reports
Alerts
Dashboards
Modules
New Search
Save As
Create Table View
Close
index=* ("useradd" OR "adduser" OR "new user")
Time range: Last 7 days
4 events (10/1/26 6:00:00.000 PM to 10/8/26 6:48:03.000 PM)
No Event Sampling
Job
Smart Mode
Events (4)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 hour per column
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
a COMMAND 1
Time
Event
10/6/26 9:48:42.696 PM
2026-10-06T17:48:42.696252-04:00 kali useradd[134445]: new user: name=labadmin, UID=1003, GID=1003, home=/home/labadmin, shell=/bin/sh, from=/dev/pts/1
host = kali source = /var/log/auth.log sourcetype = auth
10/6/26 9:48:42.695 PM
2026-10-06T17:48:42.695612-04:00 kali useradd[134445]: new group: name=labadmin, GID=1003
host = kali source = /var/log/auth.log sourcetype = auth
192.168.56.101:8000/en-US/app/search/search?q=search index%3D* ("useradd" OR "adduser" OR "new user")&earliest=-7d%40h&latest=now&display.page.search.mode=smart&dispatch.sample_r...
6:49 pm
08/10/2026

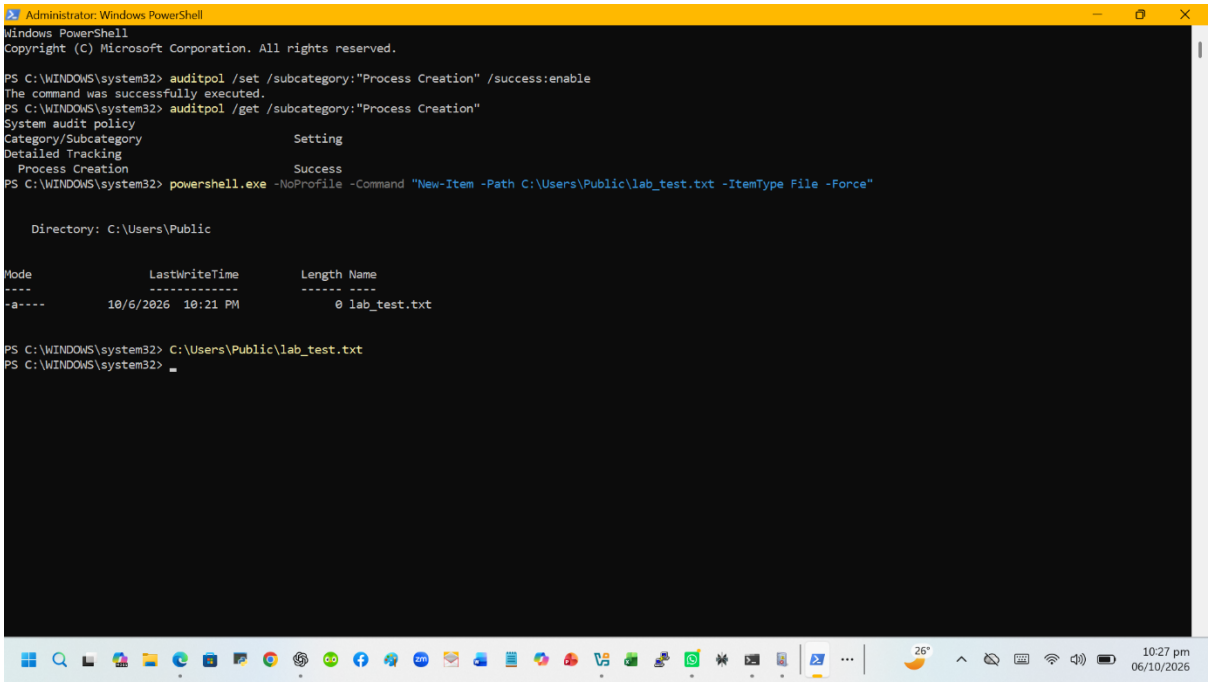

Search | Splunk 10.6.0.5
Search | Splunk 10.4.3
Not secure 192.168.56.101:8000/en-US/app/search/search?q=search%20index%3D*%20("useradd"%20OR%20"adduser"%20OR%20"new...
Sign in
Chat
Search & Reporting
Search
Datasets
Reports
Alerts
Dashboards
Modules
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
a COMMAND 1
date_hour 2
date_mday 1
date_minute 2
a date_month 1
date_second 2
a date_wday 1
date_year 1
date_zone 1
a from 1
GID 2
a home 2
a index 1
linecount 1
Time
Event
10/6/26 9:48:42.696 PM
2026-10-06T17:48:42.696252-04:00 kali useradd[134445]: new user: name=labadmin, UID=1003, GID=1003, home=/home/labadmin, shell=/bin/sh, from=/dev/pts/1
host = kali source = /var/log/auth.log sourcetype = auth
10/6/26 9:48:42.695 PM
2026-10-06T17:48:42.695612-04:00 kali useradd[134445]: new group: name=labadmin, GID=1003
host = kali source = /var/log/auth.log sourcetype = auth
10/6/26 9:48:42.595 PM
2026-10-06T17:48:42.595964-04:00 kali sudo: kali : TTY=pts/0 ; PWD=/
kali ; USER=root ; COMMAND=/usr/sbin/useradd -m labadmin
host = kali source = /var/log/auth.log sourcetype = auth
10/6/26 8:16:20.589 PM
2026-10-06T16:16:20.589758-04:00 kali useradd[66369]: new user: name=splunkfwd, UID=1002, GID=1002, home=/opt/splunkforwarder, shell=/bin/bash, from=/dev/pts/1
host = kali source = /var/log/auth.log sourcetype = auth
192.168.56.101:8000/en-US/app/search/search?q=search index%3D* ("useradd" OR "adduser" OR "new user")&earliest=-7d%40h&latest=now&display.page.search.mode=smart&dispatch.sample_r...
6:49 pm
08/10/2026

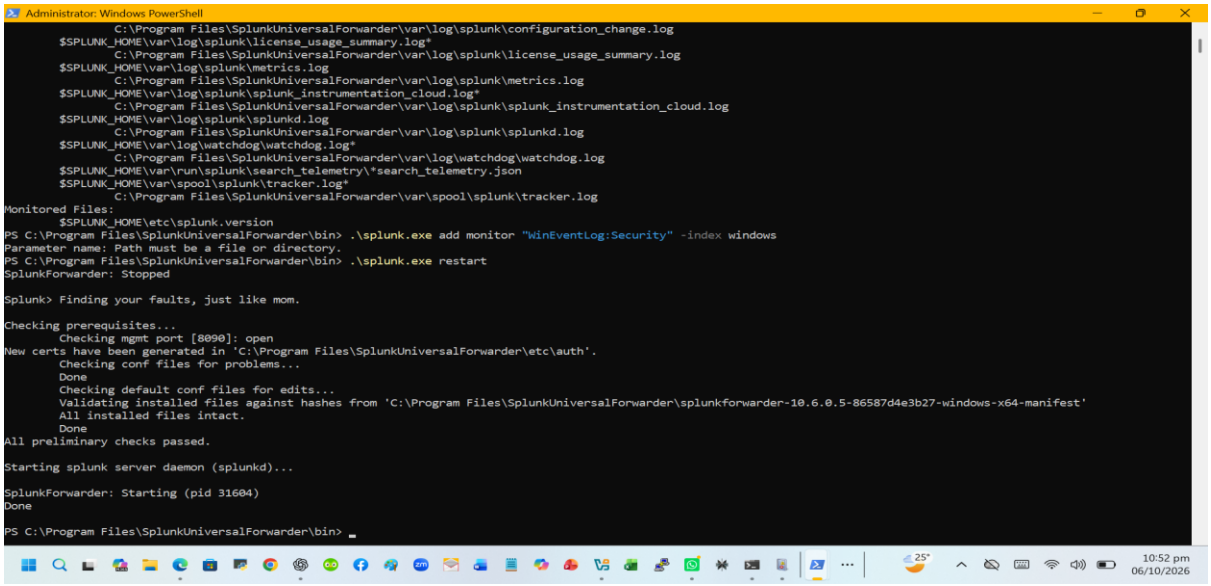
7.3 Suspicious Command / PowerShell Activity

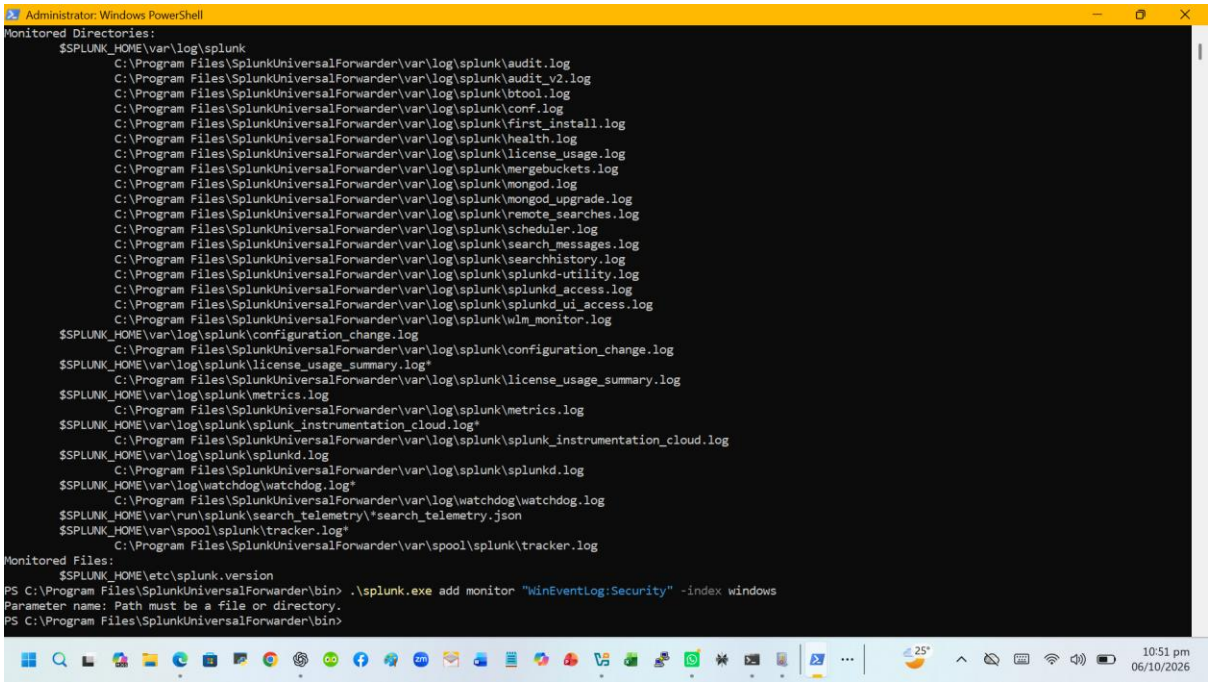
A controlled suspicious command was executed on the Windows endpoint to generate process-related security activities. Where process auditing was available, Event ID **4688** was investigated. PowerShell-related activity was also searched for in Splunk.

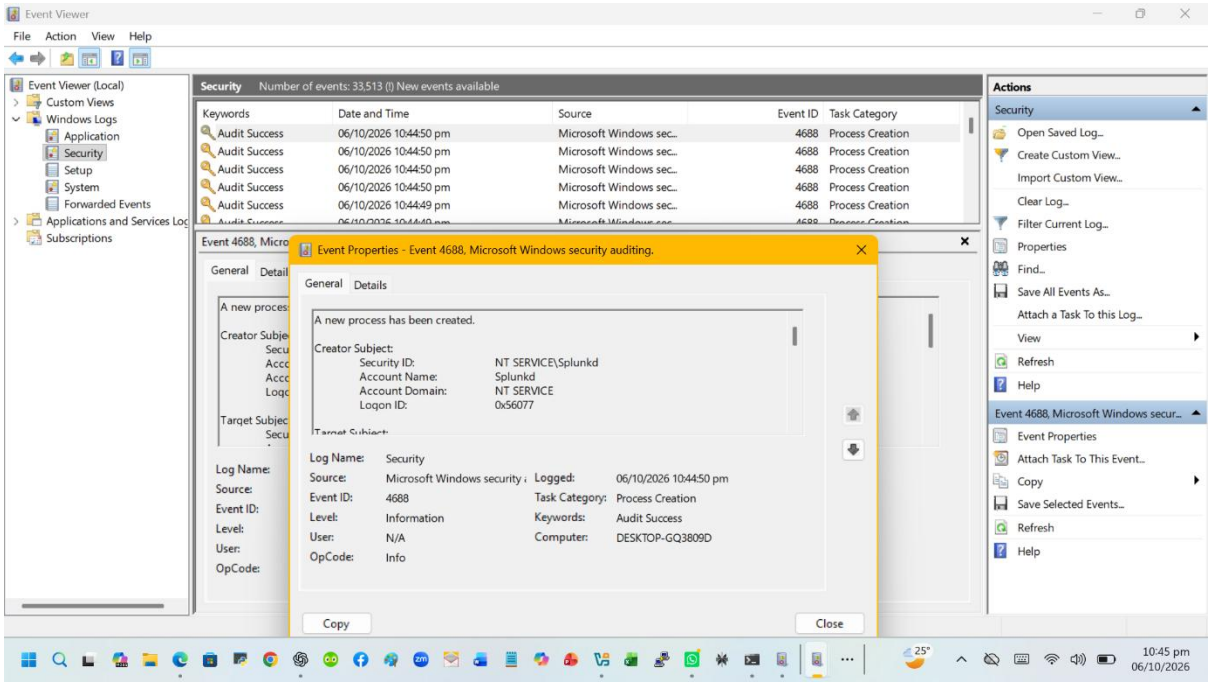
Example searches included:

index=windows source=”winEventLog:security” EventID=4688









The investigation considered the process name, user, command information, parent process, host, and timestamp.

Affected host – Window PC (DESKTOP GQ3809D)

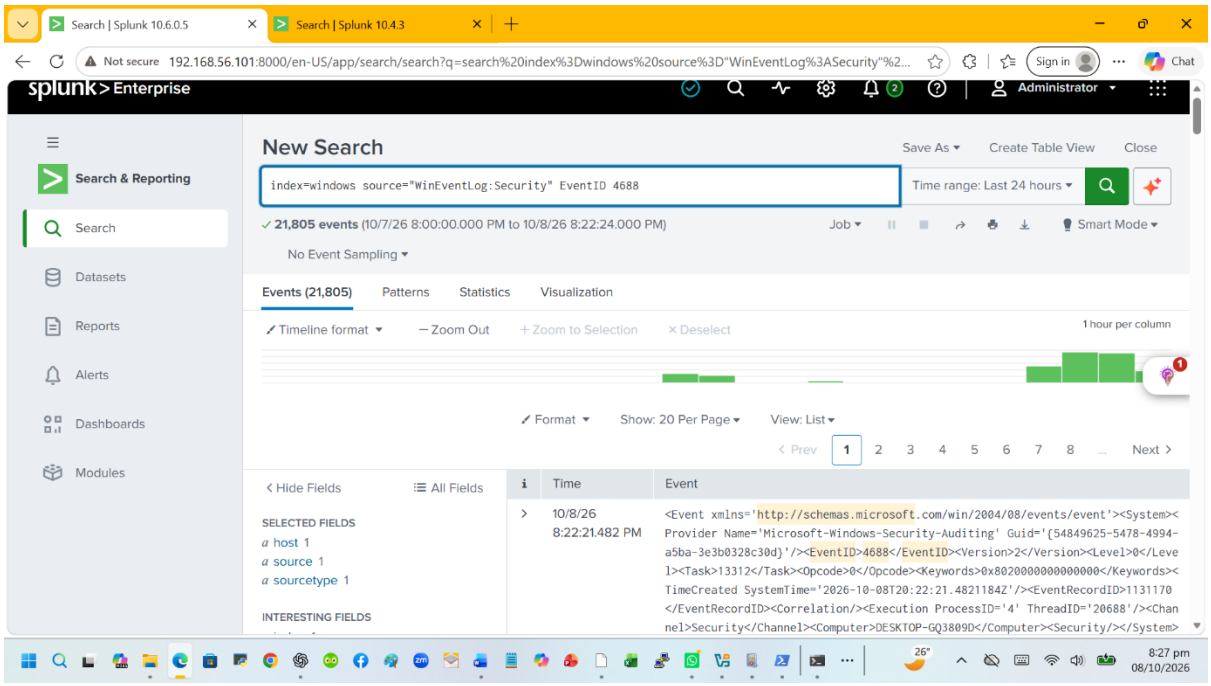
Date of the Attacks -06/10/2026

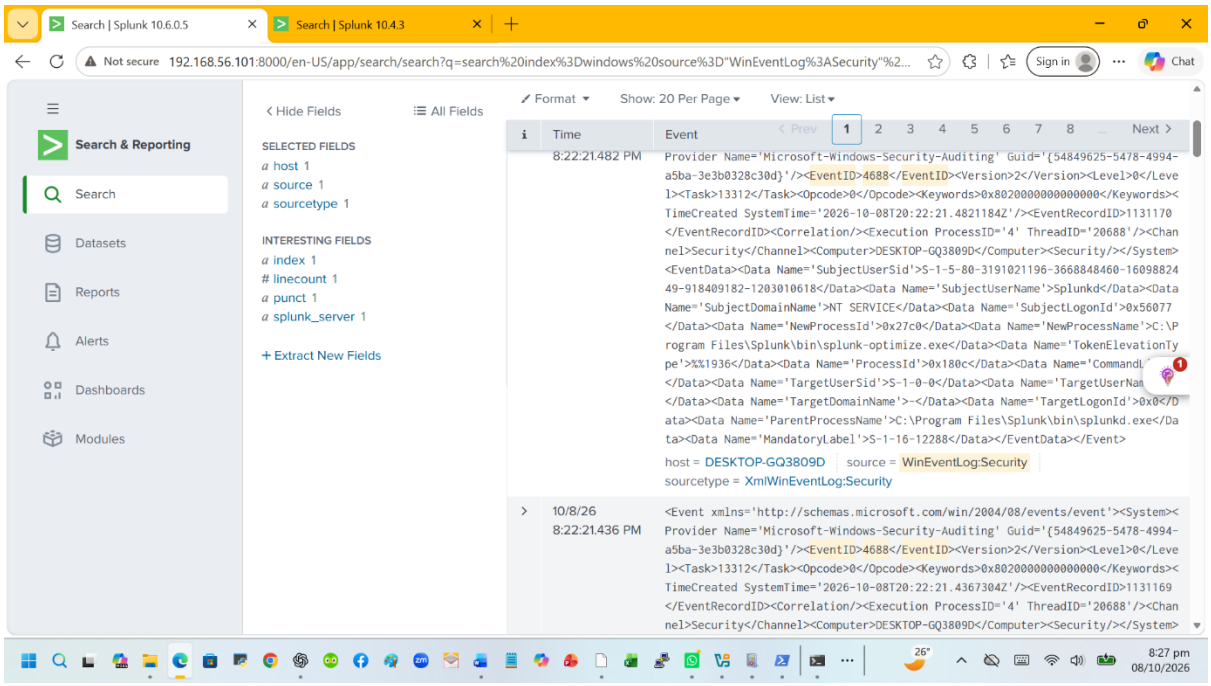
Time of attacks -10:44PM

Date and Time of Detection – 08/10/2026, 10:21PM

Process name – process creation

MITRE ATT&CK: T1059.001 – PowerShell.





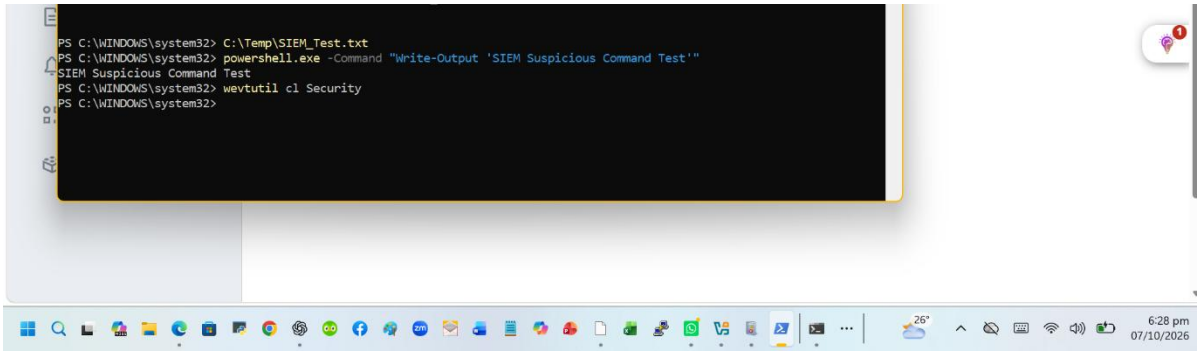
7.4 Clearing Windows Security Logs

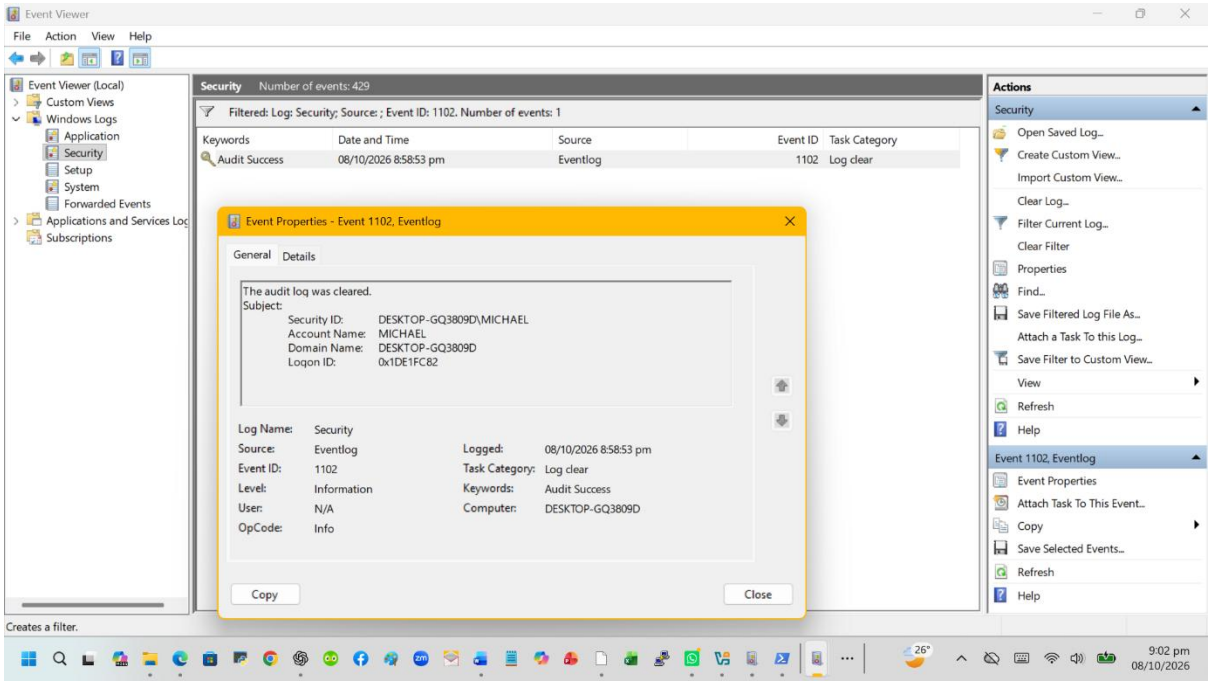
The final test simulated an attempt to remove evidence by clearing the Windows Security audit log.

Splunk was used to identify **Event ID 1102**, which indicates that the Windows Security audit log was cleared.

Example search:

index=window “1102” LogFileCleared





The event was investigated using the timestamp, affected host, user information, and event message.

This activity is important because an attacker may attempt to remove evidence after performing malicious actions

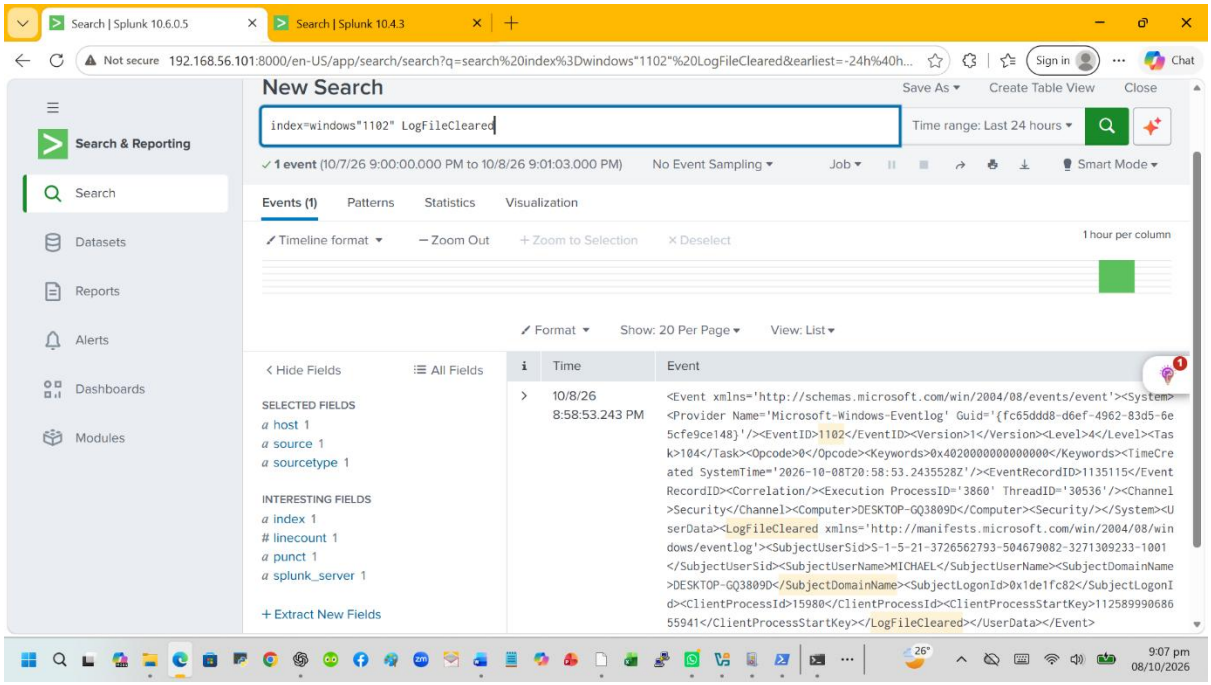
Event message – Attempts Security Logs Clearing

Affected host – Window PC (DESKTOP GQ3809D)

Date of the Attacks -08/10/2026

Time of attacks -8:58PM

MITRE ATT&CK: T1070.001 – Clear Windows Event Logs.



8. Investigation and Detection

ATTACKS	DETECTIVE EVIDENCE	SPLUNK INVESTIGATION
Brute-force	Multiple failed logins	Failed authentication events
Account creation	New user account	Linux account creation
Suspicious Commands/ PowerShell activity	Process/PowerShell activity	Event ID 4688/PowerShell
Log clearing	Security log cleared	Event ID 1102

The detective analysed the logs generated by the simulated attacks. Splunk searches were used to identify suspicious events and determine when and where the activities occurred.

The investigation helped establish a timeline showing the sequence of activities performed during the simulated incidents.

9. Incident Timeline

The following timeline has been completed using the actual timestamps from the Splunk logs:

TIME	ACTIVITY	SYSTEM	EVIDENCE
5:33PM	Brute force	Kali Linux	Multiple failed logins
5:48PM	Account creation	Kali Linux	New user account
10:44PM	Suspicious commands/PowerShell	Window PC	PowerShell activity
8:58PM	Log clearing	Window PC	Security logs cleared

10. MITRE ATT&CK Mapping

These activities were mapped to the MITRE ATT&CK Mapping framework

Activity	**Techniques**
Brute force attack	**T1110- Brute force**
Account creation	**T1136 – Create account**
PowerShell execution	**T1059.001 - PowerShell**
Clearing security logs	**T1070.001 – Clear window event logs**

This mapping helped explain how the simulated activities relate to common attacker techniques.

11. Recommendations

Based on the investigation, the following security measures are recommended:

1. **Use strong passwords and multi-factor authentication** to reduce the risk of brute-force attacks at any given time.
2. **Apply least-privilege access** so ordinary users cannot perform administrative actions.
3. **Monitor account creation** and investigate unexpected and privileged accounts.
4. **Enable PowerShell and process logging** to improve visibility into suspicious commands.
5. **Monitor Event ID 1102** and generate a high-priority alert whenever Security logs are cleared.
6. **Centralize logs in the SIEM** so attackers cannot easily remove evidence stored on individual endpoints.
7. **Regularly review Splunk alerts and dashboards** to identify suspicious activity early.

12. Conclusion

The SIEM deployment successfully demonstrated how Splunk can provide centralized security monitoring and investigation capabilities. Windows and Linux logs were collected and forwarded to the central Splunk server, allowing security events to be searched and analysed from one location.

The four simulated attacks demonstrated the importance of monitoring authentication events, account changes, process execution, and log-clearing activities. The project also showed how a SOC can use centralized logs, investigation techniques, and MITRE ATT&CK mapping to identify and respond to potential security incidents when the need arises.

Overall, the project provided practical experience in SIEM deployment, log collection, threat detection, incident investigation, and security monitoring.